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Work package E: Tracking ships

Deliverable E3: Interim technical report concerning the conditions for using the data, the methodology and the procedures

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Prepared by:

Mark van der Spoel (CBS, The Netherlands)
Remco Paulussen (CBS, The Netherlands)
Sjoerd Schenau (CBS, The Netherlands)
Tessa de Wit (CBS, The Netherlands)
Wai King Meijer-Cheung (CBS, The Netherlands)
Eleni Bisioti (ELSTAT, Greece)
Christina Pierrakou (ELSTAT, Greece)
Anna Bilska (GUS, Poland)
Michał Bis (GUS, Poland)

Work package Leader:

Remco Paulussen (CBS, The Netherlands)
e-mail address : rpln@cbs.nl
telephone : +31 (0)45 570 6853
mobile phone : +31 (0)6 8011 5907

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Executive/management summary

Building on the developments of the completed work in the ESSnet big data programme I [1] on the use of AIS data (Automatic Identification System) for maritime statistics, the aim of this Work Package (WP) tracking ships (WPE) is to develop functional production prototypes. Results of the WP in the first program showed the potential wealth of AIS data to improve current statistics and to generate new statistical products. Although some important elements of current maritime statistics such as type and quantity of goods loaded or unloaded at the port are not part of AIS, AIS is useful to improve other aspects of statistics and provide new products. In addition, AIS data is also useful for statistics on inland waterways, emissions and can provide other new products. Concerns signalled were the need for suitable hardware and software tools for the exploration and exploitation of the huge amount of AIS data and the lack of documentation and guidance for using AIS.

For each of the four main products identified in the previous deliverable, a first version of the process using AIS data has been developed in the current program:

(1) Improve statistics on Inland Waterway transport: Currently, statistics on Dutch Inland Waterways (IWW) are based on incomplete information. Information on some parts of the Dutch inland waters is missing, resulting in incomplete data on ships' journeys and with that information on goods (un)loaded. Implementing information from AIS in the process, completes this missing information on ship's journeys. Next step are to improve algorithms and estimate the type of goods ships might have carried, based on the type of terminal.

(2) Develop statistics on the behaviour of fishing vessels: Using AIS, the first results on the activity of fishing fleet (time when a fishing vessel is out of the port) and the traffic of fishing fleet (the number of fishing vessels in fishing areas) in Poland have been derived. Several visuals have been created: map with the fishing ports and fishing areas, and the traffic intensity. The fishing fleet was derived using the EU fishing fleet register, applying the rules of the Common Fisheries Policy. Next steps are to create experimental statistics and to test the solution for selected fishing ports in Greece and/or The Netherlands.

(3) Improve quality of statistics on port visits: A process workflow has been developed using AIS data to derive the port visits and port traffic of the Piraeus port in Greece. Also an animation showing the traffic of the Piraeus port for one day was created. The first results, compared to data from official statistics, showed that all passenger ships, which arrived at Piraeus Port were correctly identified (100%) and that the number of arrivals was approximately 96% correct. Using AIS data, 5 more ships were detected. These differences will be investigated.

(4) Improve statistics on air emissions and energy used: The first step of calculating the annual energy use and the air emissions from sea and coastal passenger and freight transport and fishery, is to derive all ships from Dutch shipping companies. Using text mining techniques, the WP tried to match the ships' operators in the shipping list with the shipping companies in the Dutch business register. More than 60% of the ships are not matched, so more research is needed. In addition, algorithms to calculate distances travelled on a global scale are being developed.

The products mentioned were or are being created using several big data platforms: local platforms, the Big Data Test Infrastructure (BDTI) environment provided by European Commission, and the United Nations Global Platform (UNGP) provided by the Global Working Group (GWG) on Big Data for Official Statistics. This intermediate report contains some paragraphs to share our experiences in working on those platforms.

Finally, a first version of the WPE architecture was created using the Big data REference Architecture and Layers (BREAL), as defined by the work package on architecture (WPF).

1. Introduction

Building on the developments of the completed work in the ESSnet big data programme during 2016 - 2018 [1], the aim of this work package (WP) is to develop functional production prototypes. Results of the WP in the first program showed the potential wealth of data from the Automatic Identification System (AIS), an automatic tracking system on ships for safety and management purposes. The data source contains detailed information on different aspects of the ship, such as identity, location, course, and speed, providing information to improve current statistics and to generate new statistical products. The WP consisted of three tasks: data access, data handling and methodology and techniques. It was concluded that although some important elements of current maritime statistics such as type and quantity of goods loaded or unloaded at the port are not part of AIS, AIS still is useful to improve other aspects of maritime statistics and provide new products. In addition, AIS data is also useful for statistics on inland waterways (IWW), emissions and can provide other new products. Concerns signalled were the need for suitable hardware and software tools for the exploration and exploitation of the huge amount of AIS data and the lack of documentation and guidance for using AIS.

In this second big data programme the aim is to set up procedures and develop technical solutions for the implementation of AIS for statistics on IWW, port visits, energy use and emissions. Experimental statistics on fishing fleet behaviour was also identified as an important new product. Three National Statistical Institutes (NSI's) participate in WPE: the national statistical institutes of The Netherlands (work package leader), Greece and Poland. The output of the work should become available to the European Statistical System (ESS) and enable other NSI's to produce statistical output. Therefore, active involvement of subject matter experts, statisticians, enterprise architects and other stakeholders is already needed. Subject matter experts for the NSI's working in the relevant national statistical production units (for example, maritime and IWW transport, emissions, fishing) will be involved in all phases of work. Participation to relevant Eurostat Working groups and Task Forces (for example, maritime and IWW transport statistics) will be part of the work of this WP.

Along with the production of the relevant methodologies, recommendations, specifications and statistical software, the production of experimental statistics demonstrating the capabilities to produce statistics is an objective of this WP. See "Annex 1: Scope of WPE in more detail" for a more detailed description of the work in this WP.

In this deliverable, the current status on the implementation and development of statistical products to be delivered in this WPE are described in chapter 2. This concerns improving the statistics on Inland Waterway transport, developing statistics on the behaviour of fishing vessels, improving the quality of statistics on port visits, and improving statistics on air emissions and energy used. Chapter 3 describes an experimental investigation into the possibilities of applying dynamic networks and in-port waiting times of shipping as indicators of economic trend changes. As the current WPE uses AIS data provided by another provider, EMSA, we have performed a quality analysis using the quality framework from the previous ESSnet. This analysis is described in chapter 4. Chapter 5 describes our experiences using the Big Data Test Infrastructure (BDTI) environment and the United Nations Global Platform (UNGP). These platforms are being used to develop some of our processes. In chapter 6, a first version of the WPE architecture was created using the Bigdata REference Architecture and Layers (BREAL), as defined by the work package on architecture (WPF).

2. Current state on identified processes

Inland waterways

Introduction

In statistics on the transport of goods via Inland Waterways (IWW), the amount and type of goods transported via inland waterway are described for a national territory. In the Netherlands, these statistics are based on information that has to be provided when inland waterway ships pass by a lock. At the locks information is gathered on the origin and destination of a ship, the goods the ship carries and some ship's characteristics. However, this information is known to be incomplete as not all shipping routes pass by a lock, resulting in incomplete information for these routes. Even when a ship passes by a lock, the information may still be incomplete. This project focuses on completing the missing information using AIS data.

Information sources

There are two main sources used in this project: lock information and AIS. Both sources are received from Rijkswaterstaat, part of the Dutch ministry of Infrastructure and Water Management. They are involved in the management and maintenance of the main infrastructure facilities (not restricted to waterways) in the Netherlands.

We first performed a preliminary study. The data used here comes from July 2015 and comprised of all message types, with a size of 60 GB raw NMEA coded AIS messages. Note that for IWW transport a different type of AIS message has to be used: beside message types 5, message type 8 contains static information (containing e.g. vessel information) for inland vessels specifically. For the current project we have access to 2017 data.

Concepts & Methodology

The IWW statistical process only takes journeys into account that take place on Dutch territory. A journey is defined as a movement between two locations where loading or unloading takes place, or the location where a ship crosses a national border. To estimate the missing information on journeys and traffic from AIS a method to construct journeys was developed. This is consisted of three steps: pre-processing, deriving journeys and linking AIS and traditional journey information.

Pre-processing

AIS data consists of dynamic and static messages. The dynamic messages contain information such as the ship's location and speed. The static messages contain information on the ship characteristics such as identity, size and ship type. The first step in pre-processing is decoding. After this, static and dynamic messages are separated and split into separate files for each ship.

As the dynamic information is the largest amount of data, data needs to be filtered first. This is done by selecting messages with speed faster than 0.2 knots, latitudes and longitudes in Dutch territory and a 9 digit MMSI, the ship's Maritime Mobile Service Identification number. Navigational status, signalling the activity of the ship such as "under way" or "at anchor", is not used because these are known to be incomplete or not up-to-date. Static messages are filtered to contain messages with different information. Then, a single static message is added to all dynamic messages.

Deriving journeys

After pre-processing, we need to derive journeys per ship. A journey is a movement between two locations where goods are loaded and/or unloaded or the point where a ship leaves Dutch waters. When the time interval between two successive data points is over an hour or the difference in location is larger than 200 meters, the first data point is considered to be the end of a journey and the succeeding data point the start of a new journey. Each location is linked to a register of terminal locations using a nearest node method. Using this method, the distance between the location of the ship and all terminal locations are calculated; the terminal location with the shortest distance is then selected.

As the definition requires locations where ships (un)load, the location register also has to contain locations where ships stop for other reasons beside (un)loading, such as waiting locations for locks or sleeping locations. The stopping locations will not be used to define the beginning or end of a journey.

Linking AIS and traditional journey information

Journeys from AIS (lasting from tA1 – tA2) were linked to "lock" journeys, journeys derived from the original data source, the data from locks (lasting from tB1 – tB2). The following criteria were used: the starting time of the "lock" journey had to be before the end time of the AIS journey and end time of "lock" journey had to be after beginning time of AIS journey: tA2 > tB1 and tB2 > tA1.

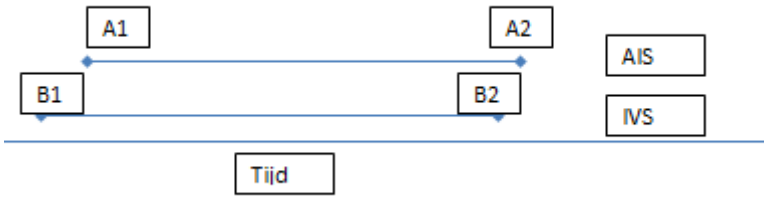


Figure 1 Scematic representation of time distribution

Then, for the "lock" journeys the amount of overlap with the AIS journey is calculated using the following link index:

$$Link\ index\ = \frac{(tA2 - tB1)}{(tA2 - tA1)} * \frac{(tB2 - tA1)}{(tB2 - tB1)} * (\mathrm{tB2} - \mathrm{tB1})$$

Equation 1 Formula of link index

If the time matches, link index will be around a value of 1, otherwise the value will lower. Values with the highest link indices are matched. AIS journeys that are not linked to "lock" journeys are taken to be missing journeys from the lock data.

We have developed several Python scripts for decoding and filtering, deriving journeys and linking. These will be published on the GitHub page for this WP. We have decided to split all steps into different scripts as this might be more useful for other institutions working with AIS data.

Preliminary results

AIS data was useful in completing information on missing journeys. Using AIS resulted in 9 percent extra journeys that were missing from the lock data. However, the journey linking algorithm was not optimal. Both false positives and false negatives occurred.

In the extra journeys from AIS data the problem also lies in the definition of a journey. The start and end of a journey should be characterized by a loading or unloading action. However, ships also have other reasons to stop, for example to rest, wait for a lock or to get fuel. Therefore, the location of a stop is used to define the occurrence of an (un)loading action. In case of a terminal, an (un)loading action is assumed. This assumption requires an accurate database of stopping locations that are functionally defined, even on a terminal level. However, this strict divide cannot always be made. The location database, in addition, is known to be inaccurate and incomplete.

Even though AIS gives us information on the missing journeys, we do not have information on the goods and quantities transported. The type of ship, the type of terminal and/or the combination of these can provide some information on the nature of the type of goods.

Implementation

After we have finished the research described, the process will be implemented. Then, data for a larger time window has to be analysed, resulting in new datasets for that specific year for all our customers Eurostat and Statistics Netherlands' StatLine). These analyses should result in a new method for merging AIS journeys and journeys from the current IWW statistics. Also, these should result in new method for deriving type and quantity of goods transported. We will also investigate whether a complete remodelling of the IWW statistics can be helpful for implementation, but this is outside the scope of this WP. Additionally, we will need to take a closer look at the hardware requirements necessary for this implementation, see "Challenges". We also need to investigate the consequences of this measure on our time-series at hand. This will probably be of great importance and pose a risk to the time schedule. The methodology department will assist with several tools.

As for implementation, a new 'Microbase' will be created out of which we can distillate the new required output. It will be a combination of SQL scripts as is the current IWW statistics, most likely combined with Python scripts for the AIS part.

Challenges

The main challenge in this project is processing the volume of the data. Each week of raw data is approximately 22 GB, making it quite cumbersome to work with within the institute. Even though infrastructure is available, it is not optimal, rendering processing time is quite long. Both infrastructural and software optimization developments are necessary.

Fishing fleet

Introduction

Behaviour of fishing fleet (activity and traffic) has not been a widely researched topic yet, however it is becoming a topic of great interest to the central and local government administration in the coastal areas. For a large part, this is a result of the sustainable development policy (see Goal 14: Conserve and sustainably use the oceans, seas and marine resources).

The analysis performed for this WP focuses on the behaviour of the fishing fleet and its activity within fishing areas, which shall be of novelty and allow production of experimental statistics on fishing fleet behaviour. The work concerning Polish fleet and selected Polish seaports should result in an easy-to-implement software with universal methods supporting the process of collecting, processing and analysing AIS data for the purposes of statistics. The results shall be tested for a single port in the Netherlands and in Greece.

An important aspect of the work is the use of available sources, namely AIS and Fleet Register data, and the fact that the developed model shall be so universal that it can be replicated for the ESS and other international organizations.

Information sources

There are two main sources used in this project: AIS data and the public register of fishing vessels. Statistics Poland receives AIS data from the Maritime Office in Gdynia, responsible for maintaining and developing the AIS infrastructure in Poland. In addition, AIS data from the European Maritime Safety Agency (EMSA) for the years 2016 and 2017 was available to WPE.

The register of fishing vessels in Poland is maintained by the Minister of Maritime Economy and Inland Navigation. The register is kept in an electronic form on the basis of a regulation of the European Commission. Each EU country is obliged to maintain (i.e. collect, update and share) its national fishing fleet register. Data from the EU fishing fleet register are used to apply the rules of the Common Fisheries Policy. According to the Commission Implementing Regulation (EU) 2017/218, technical data on individual fishing fleets are published on the website

http://ec.europa.eu/fisheries/fleet/index.cfm?lg=en

Based on the public register, we prepared a detailed database of the Polish fishing fleet. It helps us to generate an up-to-date reference frame of fishing fleet, enabling filter AIS data in order to reduce the amount of data to be processed.

Data comparison – EMSA versus national Polish AIS data

In order to compare the quality of the national AIS data (AIS-PL) with European AIS data from EMSA, both data sources were joined. This comparison was prepared performed for January 1-2 2016 for the port of Świnoujście, using the same codes and approach from the previous SGA work

Day of measurement	2016-01-01		2016-01-02	
	Number of distinct ships			
AIS-PL	12		15	
EMSA	11		16	
Common data	11		15	
	Number of appearances for common data			
Numer of ship	AIS-PL	EMSA	AIS-PL	EMSA
1	129	9	130	7
2	123	7	126	5
3	679	72	646	62
4	149	8	159	15
5	122	9	138	8
6	188	8	40	7
7	1785	100	257	10
8	76	6	24	9
9	134	6	123	7
10	120	7	171	10
11	362	18	138	9
12			157	6
13			139	14
14			51	12
15			678	61
quantitative comparison - dynamic data (1,2,3)				
Day of measurement	2016-01-01		2016-01-02	
AIS-PL	EMSA	AIS-PL	EMSA	AIS-PL
count(*) - records	1311107	1424757	1322558	1444895

Figure – the port of Świnoujście

Name of port	Port of Świnoujście
Point A (latitude, longitude)	53.951968, 14.286217
Point B (latitude, longitude)	53.907008, 14.250708

Figure 2 Comparison of national Polish AIS data (AIS-PL) with EMSA for the port of Świnoujście

Both sources showed almost the same number of unique vessels for particular dates (see Figure 2). Regardless of which of those sources was analysed, the difference in the number of ships was one ship. However, there are discrepancies in the number of messages sent by the ships in the datasets. See the number of appearances, where the first two columns show the number of messages per ship on 1. January 2016 in the sources AIS-PL and EMSA respectively. The last two columns show the number of messages per ship on 2. January 2016. Although the number of ships contained in the EMSA data (for 2 January 2016) was larger, the AIS-PL data contains a significantly larger number of messages. The smaller amount of messages probably resulted from data filtering – the frequency of the EMSA data seemed to be filtered to a 6 minute interval.

We concluded from this that the national data are more precise (due to their dense frequency) than the EMSA data. Depending on the goal of using the data, EMSA data might be preferable due to a lower amount of data. However, for the goal of investigating fishing behaviour, more detailed information is needed. Thus, in order to prepare a prototype experimental statistics on traffic and activity of the maritime fishing fleet, we will use the AIS-PL data. However, to test the algorithms for a selected fishing port from the Netherlands and Greece, we will use the AIS data from EMSA.

Concepts & Methodology

Our approach to develop experimental statistics on the behaviour of fishing vessels with the use of AIS data and the PL fishing fleet register is presented in the scheme below (Figure 3).

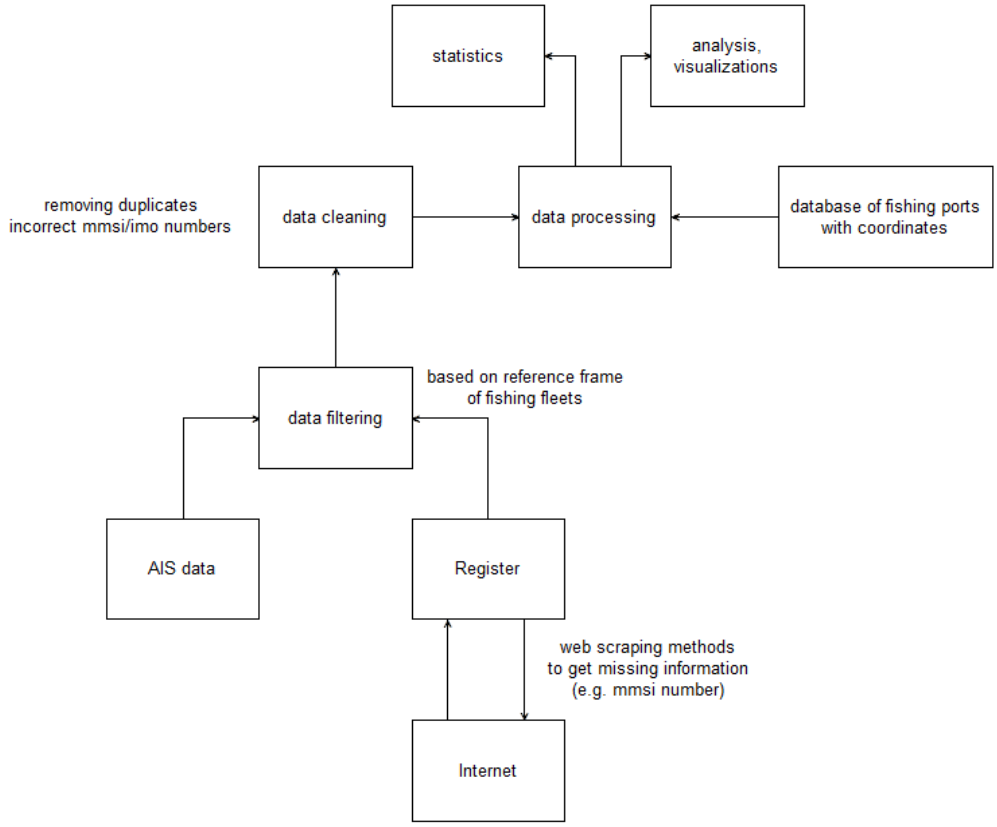


Figure 3 Schematic diagram – fishing fleet experimental statistics

AIS data contains messages from all types of ships ("AIS data" in Figure 3). In our case-study on Polish fishing fleet statistics we also needed to select Polish fishing vessels only. To do that, we started with preparing a reference frame of Polish fishing fleets.

As mentioned, every EU country is obligated to report, maintain and update its country's fishing fleet register in unified format. Eurostat's database stores this information and we downloaded the Polish dataset of fishing vessels. Unfortunately, as it turned out there was a lot of information missing (for example incorrect/missing MMSI-number, IMO-number) which will make it difficult to connect our reference frame to AIS data (for example through MMSI or IMO). To update/complete information that was missing we implemented Python web scraping methods to selected web pages (for example fleetmon.com or vesselfinder.com) contain ship data. Information from web scraping was compared to Register data and fleet dataset was updated (see Figure 4).

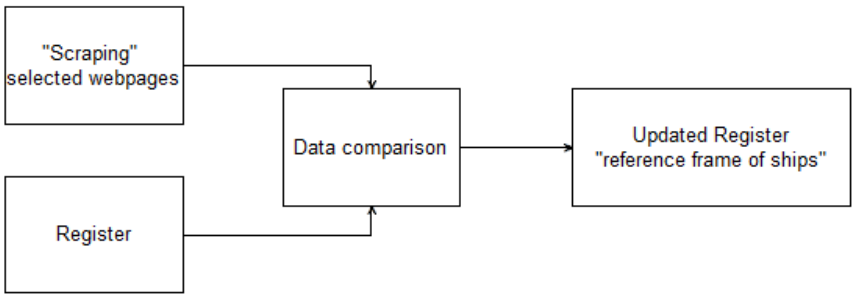


Figure 4 Schematic diagram – supplementing incorrect/missing information via web scraping methods

After supplementing incorrect/missing information we created a reference frame of Polish fishing fleets ("Register") and were able to filter AIS data ("data filtering"), thus limiting our AIS dataset. We simply removed all information from other ships.

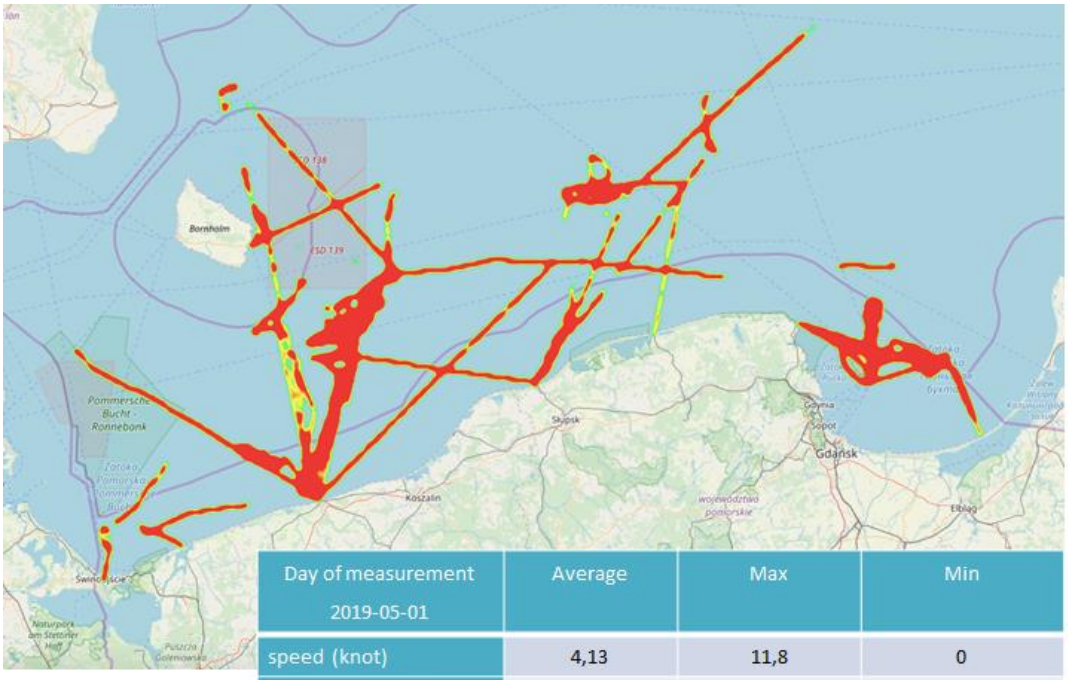
The next step with the remaining data was to remove (data cleaning) outliers (for example, coordinates showing positions on land), technically faulty messages (for example, due to data noise) and messages that were duplicated. The result of this was a clean AIS dataset concerning the Polish fishing fleet only.

What is specific for fishery is that usually each fishing boat is "assigned" to one fishing port. To be able to measure and analyse activity and traffic of fishing ships, we needed to set up a database of fishing ports with its geographical coordinates (rectangular shape of area). This kind of data is also included in Eurostat's datasets that we downloaded at the beginning. Combining ports' coordinates and ships' AIS messages together we are able to prepare code that helped us to analyse, for example, when, how many times a day, for how long a fishing vessel is active or in the home port. At the moment, we are still developing algorithms that will allow us to measure fishing behaviour outside the port, at sea. Visualising geospatial information using heat maps can help us distinguish areas of unusual (or regular) fishing activity.

Results

The first results on the activity of fishing fleet (time when a fishing vessel is out of the port) and the traffic of fishing fleet (the number of fishing vessels in fishing areas) in Poland have been derived.

Several visuals have been created: a map with the fishing ports and fishing areas, and the traffic intensity. It is possible to differentiate between itineraries typical for fishing vessels on the basis of AIS. Routes involving fishing activity take the shape of zigzags, while the routes travelled to reach a fishery are generally straight lines. When fishing, a fishing vessel sails at a slow speed, but quite often changes the course.



Day of measurement 2019-05-01	Average	Max	Min
speed (knot)	4,13	11,8	0
draught (metres)	2,9	4,2	1,1

Figure 5 Traffic intensity of fishing fleet in Poland - 2019-05-01

Implementation

The implementation of the solution will be prepared for the Hadoop environment by using Apache Spark for data processing. For this purpose, algorithms in Scala/Python programming languages will be prepared. Data visualizations will be made available in the R language. In addition, we also intend to prepare our solution in an alternative environment - Elastic Stack: Elasticsearch, Kibana.

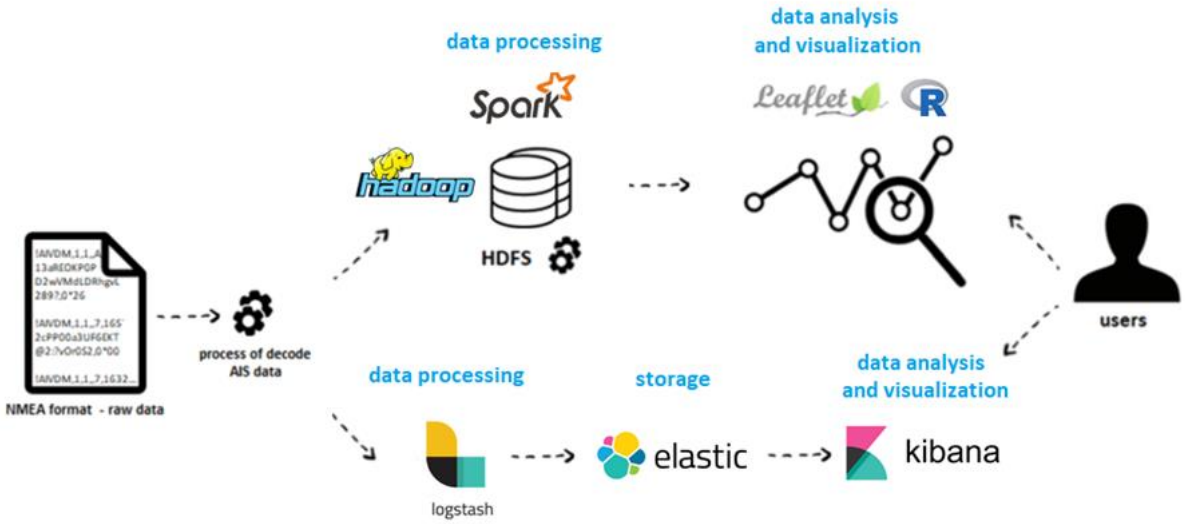


Figure 6 Overview of dataflow for AIS data - use-case for fishing fleets

Challenges

It is feasible to identify the activity of a particular fishing vessel through analysing the route velocity histogram (where a high speed means navigating along the route, a low speed means fishing activity). The distribution of the speed characteristics for particular ships might be analysed through visualization. When analysing Polish fishing fleet behaviour, there is a challenge involving the development of algorithms that allow to detect the activity of the abovementioned fleet.

Other factors related to fishing behaviour that have to be investigated:

1. Weather conditions – wind force and direction – one of the most crucial factor that affects velocity
2. Fishing license validity – affects the possibility of running the fishing activity
3. Type of fishing gear – affects velocity and draught of a ship
4. Velocity – is used to define whether a ship is actively fishing

Based on the AIS data, it is possible to define routes typical for fishing vessels. The journeys involving fishing activity form zigzags, whereas routes involving reaching the fishing area usually form straight lines since a fishing vessel sails at a slow speed when fishing, and quite often changes course. When fishing, the velocity changes unpredictably and quickly.

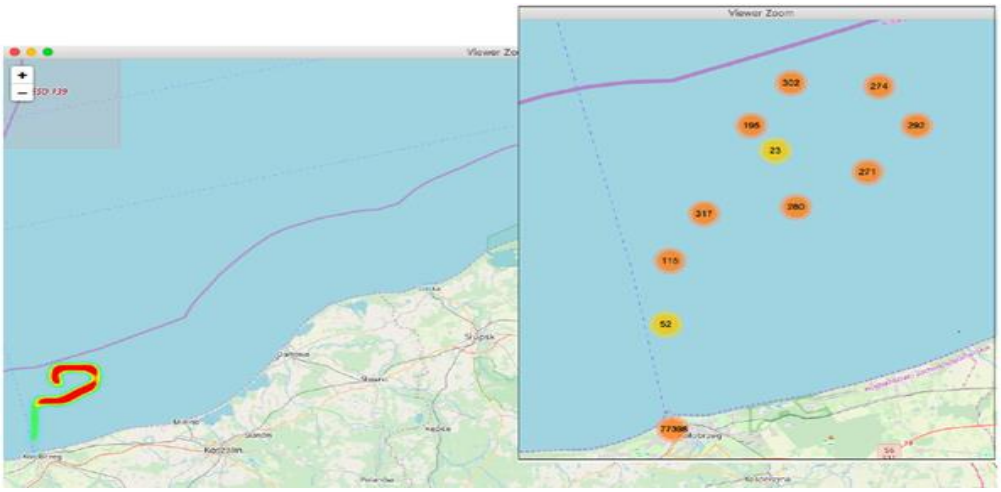


Figure 7 Analysing fishing vessel's route and movement patterns

Next steps

1. Preparation of codes to obtain / improve:
   - experimental statistics on activity and traffic of the fishing fleet
     - activity:
       - Number of active fishing vessels for a particular port in a defined time;
       - Number of vessels at a particular port;
       - Number of expected arrivals in seaports by fishing vessels;
     - traffic of fishing vessels
       - Average (the maximum and minimum) time of activity of the fishing fleets depending on the season and/or weather conditions;
       - Average (the maximum and minimum) distances travelled by a fishing vessel depending on the season and/or weather conditions;
       - Average (the maximum and minimum) velocity/draught of the fishing fleet for specific fishing areas and a time;
       - traffic / intensity of the fishing fleet in a particular area;
     - fishing capacity factor
   - detection of fishing behaviour
2. Add information about the weather conditions
3. Improve visualization methods
4. Evaluation and description of preliminary results
5. Testing the solution for other fishing port outside Poland(NL, GR)
6. Preparation of the guide for users (Starter Kit)
7. Share code on Github: https://github.com/AIS-data/WPE_Fishing_Fleet

Validation of port visits

The primary aim of this prototype is to use AIS data for the compilation and verification of existing Maritime Transport Statistics (MTS) under the Directive 2009/42/EC. Each member state collects data on loading/unloading of goods and passengers by seagoing vessels docking in their territorial ports. This data is processed into the so-called F2-table, consisting of a dataset on ships with a gross tonnage of 100 and more, visiting ports to load or unload cargo in the main European ports. The number of visits at port level are broken down into port, type and size of vessels (loading or unloading cargo, embarking or disembarking passengers including cruise passengers on cruise passenger excursion). The F2-table is mandatory for "main ports". Smaller ports can report for this dataset on a voluntary basis.

The port of Piraeus, which was selected in our case, is one of the busiest ports in Mediterranean with many terminals (Figure 8). This role is largely due to the central location, as it is the first European westbound port after crossing the Suez Canal, combined with the suitable infrastructure to serve international trade, according to the Piraeus Port Authority [2].

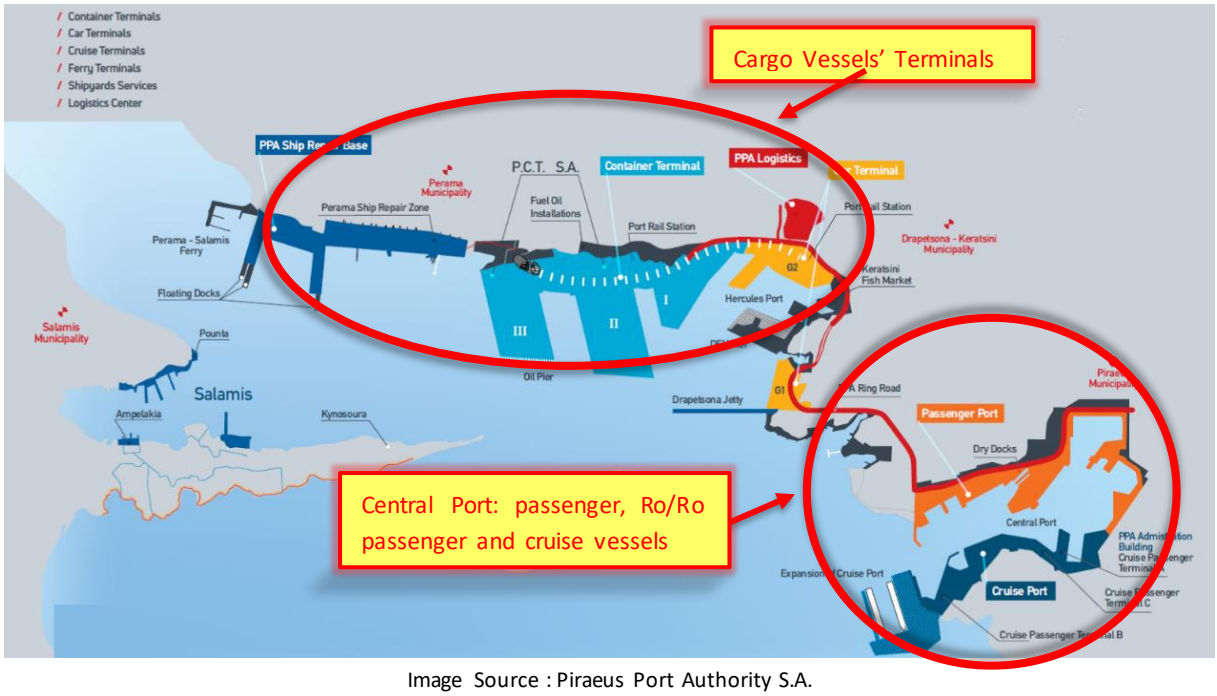


Image Source : Piraeus Port Authority S.A.

Figure 8 Terminals of the Piraeus Port

After selecting the port, the knowledge of the locations of the terminals, where different types of vessels approach the port is essential. In Piraeus port, passenger, Ro/Ro passenger and cruise vessels arrive at and depart from the Central Port (Figure 8). Other types of ships load or unload cargo at other terminals depending on ship's and cargo's type (liquid, dry bulk, containers, vehicles, etc.) (Figure 8).

Information sources

Available sources

The sources used are the following: European AIS data, available for WPE from the European Maritime Safety Agency (EMSA) for the years 2016 and 2017 and national AIS data from the Hellenic Cost Guard (HCG) available for a 6 months period (8/10/2015 – 12/04/2016) for Greece from the previous ESSnet BD I. The data thus only overlapped for the 2016 period, however this was sufficient for comparing the quality of the datasets (based on a typical day, March 30th, 2016), designing the port visits process and calculating the first results.

We also used the Ships Register from IHS Markit (Lloyds Register of Shipping), containing detailed ship data such us ship type and size (gross tonnage), which are needed for the current process. Note that the IHS Markit's Ships Register data available to ELSTAT refers to ships with at least one kind of Greek relation.

Vessel unique identifiers – Data sources' linkage

Maritime vessels can be uniquely identified by IMO (International Maritime Organization) number or MMSI (Maritime Mobile Service Identity) number. IMO number is assigned to a vessel on its birth and is permanent until its death. MMSI is also a unique identifier but might be changed if there is a change in ownership.

AIS data consist of different messages containing both dynamic (vessel position reports by message types 1, 2 and 3) and static (vessel characteristics, by message type 5) information. AIS vessel position reports contain the MMSI as identifier. AIS vessel static information contains both MMSI and IMO numbers. IHS Markit's Ships Register dataset contains IMO number as identifier. Data sources' linkage is shown in Figure 9.

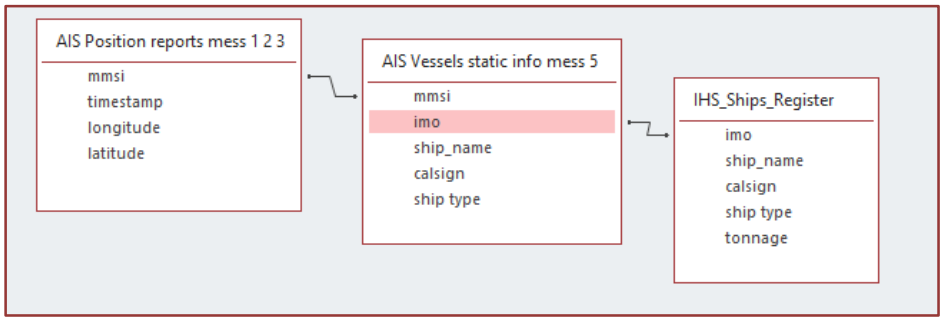


Figure 9 Data sources linkage

Preparation of Ships Register dataset

Although AIS vessel's static information (message 5) contains details about ship type and size of vessel (vessel's dimensions), it is not as detailed as required to compile the F2-table, so vessel data from IHS Markit's Ships Register was used. As described in the Directive 2009/42/EC of Maritime Transport Statistics, the type of ship in the F2-table is based on the International Classification of Ships by Type (ICST-COM). IHS Markit's Ships Register contains a more detailed classification of ship types. These had to be reclassified to the ICST-COM classification. In Table 1, the reclassification of ship type is presented. On the left side, the ship types and codes of the ICST-COM are shown. On the right side, the IHS ship types are shown and how these map to ICST-COM codes.

TYPE OF SHIP CLASSIFICATION (ICST-COM)			IHS SHIP TYPES	Assigned code
Code	*Type*	*Ship categories included*		
10	Liquid bulk	Oil tanker Chemical tanker LG tanker Tanker barge Other tanker	Asphalt/Bitumen Tanker Bunkering Tanker Chemical Tanker Chemical/Products Tanker Crude Oil Tanker Crude/Oil Products Tanker LNG Tanker LPG Tanker Products Tanker Shuttle Tanker Tanker (unspecified) Water Tanker ...	10
20	Dry bulk	Bulk/oil carrier Bulk carrier	Bulk Carrier Bulk Cement Storage Ship Cement Carrier Limestone Carrier ...	20
31	Container	Full container	Container Ship (Fully Cellular) ...	31
32	Specialised	Chemical carrier Irradiated fuel Livestock carrier Vehicle carrier Other specialised	Effluent carrier Heavy Load Carrier Livestock Carrier Vehicles Carrier Waste Disposal Vessel ...	32
33	General cargo, non-specialised	Reefer Ro-ro passenger Ro-ro container Other ro-ro cargo Combination carrier general cargo/passenger Combination carrier general cargo/container Single-decker Multi-decker	Air Cushion Vehicle Passenger General Cargo Ship General Cargo Ship (with Ro-Ro facility) Open Hatch Cargo Ship Palletised Cargo Ship Passenger/Ro-Ro Ship (Vehicles) Passenger/Ro-Ro Ship (Vehicles/Rail) Refrigerated Cargo Ship Ro-Ro Cargo Ship ...	33
34	Dry cargo barge	Deck barge Hopper barge Lash-seabee barge Open dry cargo barge Covered dry cargo barge Other dry cargo barge nes	Hopper, Motor Hopper/Dredger (unspecified) propelled ...	34
35	Passenger	Passenger (excluding cruise passengers)	Passenger Ship	35
36	Cruise Passenger	Cruise ships only	Passenger/Cruise	36

Table 1 Reclassification of vessel's types

Also, gross tonnage size classes were added to the dataset. This resulted in a Register of Vessels csv file (Ships-Reg) that contains the information needed under the Directive 2009/42/EC.

Quality aspects of AIS dataset

Expanding the validation checks under the previous ESSnet Big Data, the European AIS dataset from EMSA was compared to AIS data from Hellenic Coast Guard (HCG) , in order to examine their quality. Since overlapping data was available only for 2016 and not for the 2015, which was compared in the previous ESSnet, the comparison was performed for a typical day, March 30th, 2016[1]. We ran the checks again for the same rectangle areas defined in the previous ESSnet BD I project [5]. So, AIS was analysed in terms of the number of ships tracked by each dataset across the whole country of Greece and for the five main ports (Piraeus, Thessaloniki, Volos, Patras and Heraklion).

[1] The selected day 30/3/2016 is typical, because there were no unusual events that day, such as a national holiday, strike, etc. Moreover, all the maritime vessels in the defined rectangle areas were counted and it is not estimated using any kind of model in order to worry about seasonality.

Day of measurement: 30/3/2016

		Ports				
Number of Ships	Greece	Piraeus	Thessaloniki	Volos	Patras	Heraklion
AIS -HCG	1373	261	48	35	16	7
AIS-EMSA	1393	266	48	35	16	7

Table 2 Number of vessels tracked by HCG and EMSA data sources

The results, presented in Table 2, are almost the same. Small differences in the number of vessels tracked for Greece and for the Piraeus port may have occurred during the decoding or cleaning phase of some raw messages of the HCG.

Then the number of vessels and the number of messages per hour Greece were also compared for the two sources. The results are presented in Figure 10 and Figure 11 respectively.

Comparison between the two data sources in Greece rectangular area, 30/3/2016

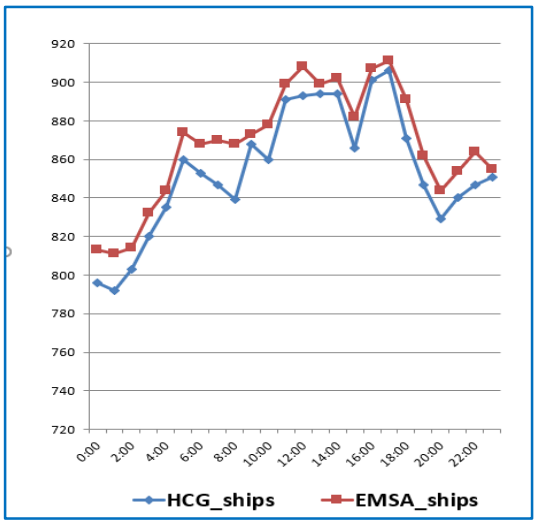


Figure 10 Number of vessels per hour

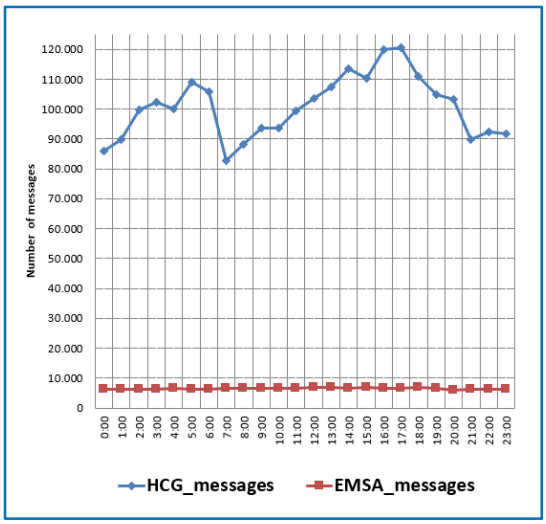


Figure 11 Number of messages per hour

The results presented in Figure 10 and Figure 11 show that the number of messages from EMSA dataset was much lower. EMSA receives from the member states messages down-sampled to 6 minutes, however more information on this is needed.

For this reason, an extra check concerning EMSA dataset was performed for Greece on 30/3/2017. The results, presented in the Figure 12, showed the number of ships and the messages from EMSA's AIS dataset as well as the expected number of messages every six minutes, per hour. So, the messages are down-sampled to more than 6 minutes.

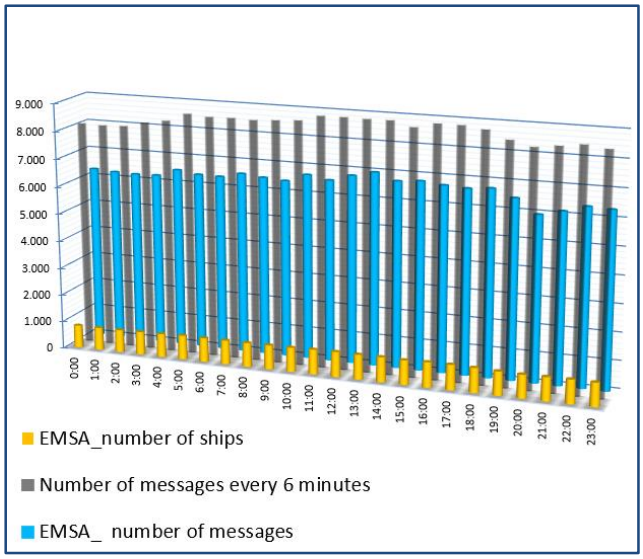


Figure 12 Number of ships, messages and expected messages every 6 minutes per hour, 30/3/2016

Pre – processing

Download AIS data from BDTI

Decoded AIS vessel position reports (message types 1, 2 and 3) and AIS vessel characteristics (message type 5) were saved in separate folders in BDTI platform. Each file covers European AIS messages for one hour. There are 24 files for each day for AIS vessel position reports and the same number of files for AIS vessel characteristics. Using the necessary credentials given to us as members of WPE, we connected to BDTI platform and after running batch scripts via AWS (Amazon Web Services) we downloaded files for one month from 1/3/2017 to 31/3/2017. In terms of volume this means 744 csv files of AIS vessel position reports for March 2017 contained 115,619,544 messages and 744 csv files of AIS vessel characteristics contained 87,347,127 messages for all Europe.

Deduplication of AIS vessel characteristics information

As far as the AIS vessel characteristics (message type 5) files is concerned, we have to deal with a volume of 87,347,127 rows needed to be de-duplicated, since the same information (vessels MMSI number, IMO number, type, name, callsign, etc.) is transmitted by each vessel's antenna every 6 minutes. In this case, we attempted first to de-duplicate the rows based on all variables. The result gave us a csv file with 27,834 rows. Then, we continued the process as each MMSI number should be referred only once, being the key variable for the data linkage. If multiple MMSI – IMO couples occurred we kept only MMSI with highest number of IMO counts. This resulted to csv file of 14,974 rows.

Methodology and preliminary results for the development of the prototype

Create AIS Geo-Database for Piraeus port area

We created an AIS database with PostgreSQL which is available under an open source license. It has spatial capabilities by means of an extension, called PostGIS. This extension allows to use spatial data types, spatial indexing and offers spatial predicates (for example to determine whether a point is inside a polygon).

The vessel position reports csv files (115,619,544 rows-messages) were uploaded to a Position Reports Table, named T-Movements. We reduced the volume of the table by keeping only messages (668,797 records-messages) from a rectangular area including Piraeus Port (Figure 13)

Figure 13 The Piraeus Port rectangle of interest

The T-Movements, which describes the activity of ships, contains MMSI number, timestamp, longitude, latitude for each vessel. From T-Movements, a list of unique MMSI numbers was derived, named T-MMSI-active, providing us with the identity of ships involved.

The MMSI-IMO couples file was imported to the database, named L-Connect, as well as the already prepared data from IHS Markit's Ship Register, named Ships-Reg. Filtering this table and keeping only records linked to L-Connect (by IMO number) a Register of ships named T-Register, for all the vessels was built up included in T-MMSI-active. So, T-Register contains IMO number, MMSI number, name of vessel, type and gross tonnage size class of vessels

<u>Visualization</u>

AIS, as a rich spatiotemporal dataset, provides the opportunity to be visualized using appropriate tools (ESRI ArcGIS, QGIS, Python, R, etc.) for better understanding the movements, also allowing the validation of calculations (whether ships are properly located spatially, whether ship density is as expected, etc.). A visualisation example of T-Movements using QGIS is depicted in Figure 14.

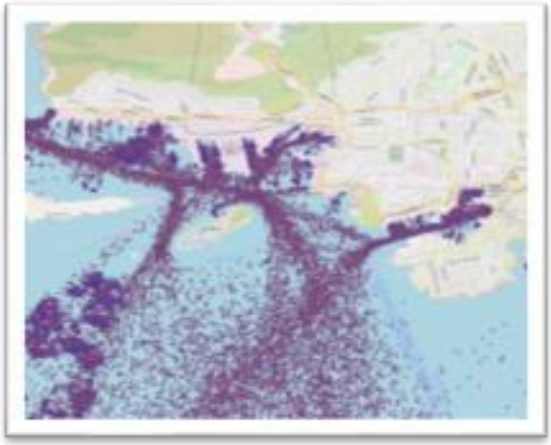


Figure 14 The Piraeus Port AIS position reports, March 2017

Split the port visits process in two separate parts

Although the methodology follows similar steps, we separated the port visits process in two parts due to the different locations of the terminals where different types of vessels approach the port of Piraeus, as it is depicted in figure 2. More specific, the Part1 focus on Piraeus Central Port where passenger, Ro/Ro passenger and cruise vessels arrive, while the Part2 focus on Piraeus Cargo terminals for Cargo vessels. We split T-Movements table to T-PassMovements and T-CargoMovements tables accordingly, by filtering the corresponding types of ships for Part1 and Part2

Part 1: Calculation of ships entering Piraeus Central Port

The T-PassMovements table was enriched with fields showing previous and next position of each ship, the time the ship needed to travel between the two positions (duration_secs) and distance covered by the ship from position1 to position2 (meters).

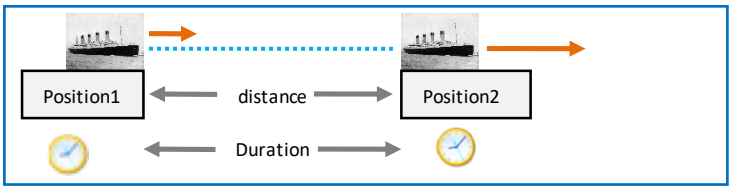


Figure 15 Enriching the passenger AIS movements table

Basic cleaning of T-PassMovements table was necessary. Rows that had a distance between the previous and next ship's position equal to zero, were deleted.

When a ship approaches to enter the Central Port of Piraeus the value of latitude from position1 to position2 increases (Figure 16). The opposite happens when a ship leaves the port. Rows satisfying criterion latitude2 > latitude1 between previous (position1) and next position (position2) of a ship were selected.

Figure 16 As the ship approaching to the Central Port of Piraeus entrance latitude is increasing

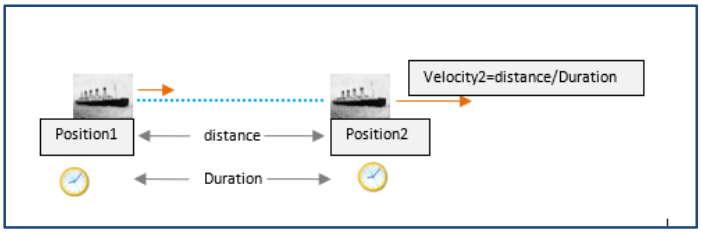


Figure 17 Calculating the velocity of the ship

The reduction of the ship's velocity arriving at the port consists another criterion, that it is used. More specific, the velocity of the current position (Position2) was calculated by dividing the distance between previous (Position1) and current ship's position to the time needed to cover that distance (Duration).

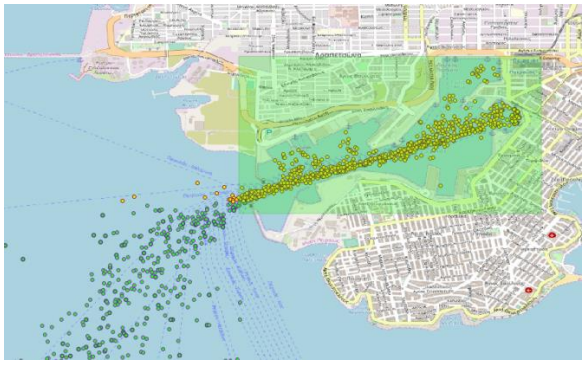


Figure 18 Applying Geo-filter to ships' position reports

At last, a geo-filter was used to separate ships' positions outside the Central port from those inside the port. Finally, the calculation of arrivals for each vessel, per day for the whole month, was conducted.

Preliminary Results

The first results compared to data from Maritime Statistics Department showed that all passenger ships which arrived at Piraeus Port were correctly identified (100%) and the number of arrivals was approximately 96% correct. Using AIS data 5 more ships were detected. The differences are to be checked.

Next steps

Part 1 of the process gave us very encouraging results. We will continue Part2 of the process, where we focus on Piraeus Cargo terminals for Cargo vessels. We are going to check the differences in Part 1 derived from the above-mentioned comparison and proceed with the compilation of F2 table. We are also going to put the code used on GitHub.

Energy used and air emissions of shipping vessels for the environmental accounts

In order to reduce climate change, increase green growth and resource productivity many national and international agreements and regulations have been formulated. In order to help governments to achieve their objectives, measurements on the origin and amount of the emitted greenhouse gasses and energy used are important. In the statistical system Environmental Accounts this information is

available and the data on the emitted greenhouse gasses are reported in the Air Emission Accounts (AEA) and the energy use in the Physical Energy Flow Accounts (PEFA).

The Environmental accounts are defined as:

> "*The Environmental accounts are a statistical system bringing together economic and environmental information in a common framework to measure the contribution of the environment to the economy and the impact of the economy on the environment. They offer a means of monitoring the pressures exerted by the economy on the environment and of exploring how these might be abated. Environmental accounts organise environmental data from many domains using the same concepts and terminology as the national accounts, and thus they show the interaction between economic, household and environmental factors and consequently are more informative than national accounts alone.* "
>
> (Source: Eurostat, 2019)

The Air Emission Accounts (AEA [7]) and Physical Energy Flow Accounts (PEFA [6]) are modules of the European Environmental Accounts (Regulation (EU) No. 691/2011) and contribute directly to the Union's environmental policy priorities. The AEA and PEFA are compiled with the use of the same concepts and terminology as the international UN System of National Accounts (SNA) and its European version (ESA). Due to the same common framework as the system of national accounts, the impact of the economy on the environment can be directly derived.

In the AEA and PEFA, the air emissions and energy used for international shipping are compiled according to the residents[2] principle of the National Accounts. In the Netherlands currently no high quality data is available for these processes with regard to energy use and air emissions in the area outside of the Dutch territory. This data is needed to complete and improve the compilation of the total air emissions and total energy used by the Dutch economy. By using Automatic Identification Systems (AIS) data the distance travelled by international shipping outside of Dutch territory becomes available.

The aim of this feasibility study is to research whether AIS data can be used for compiling energy use (for the PEFA) and air emissions (for the AEA) by shipping vessels for the environmental accounts. The research will be divided in three parts:

1. The first part is to develop a method to identify shipping vessels that are related to the Dutch economy. Shipping vessels have different types of characteristics that are linked to a nation, for example the ship flag or register owner. The question is, which of the characteristics relates to the shipping vessel contributing added value to a national economy.
2. The second part is to investigate whether it is possible to retrieve and implement worldwide AIS data for Dutch shipping vessels to calculate travel distances. Following the principles of the national accounts, all production from residents has to be incorporated, therefore worldwide AIS

[2] "The national accounts concept of residence is based on the following principle: an economic unit is said to be a resident unit of a country when it has a centre of economic interest in the economic territory of that country, that is, when it engages for an extended period (1 year or more) in economic activities in that territory" (Source: Section 5, *Eurostat Manual for air emissions accounts, 2015 edition*)

data coverage is needed. Part of the method to calculate shipping distance from AIS data was already developed in the previous ESSnet on AIS. These results will therefore be incorporated in this second part.

3. Finally, the third part is to analyse how to compile energy use (for the PEFA) and air emissions (for the AEA) based on the results from part 2: the travel distance from AIS data of residence shipping vessels. This will be done by applying the appropriate specific energy use factors (for example, energy use / kilometre) and emissions factors for example, emissions / energy use).

The study will not only focus on residents' ships, but also look at shipping vessels sailing under Dutch flag. This latter selection is the reference group for some Dutch marine policies. Also, for some EU countries this selection is the reference group for compiling air emissions. Thus, this extension in the research might be useful for those countries as well. We will compare the results for the two groups.

In addition, the research will be elaborated with fishery boats and vessels for the fishery sector. Most probably the same method of shipping vessels from part two and three, can be implemented for the fishery boats. Therefore a short study will be conducted on this subject as well.

In this deliverable results from part 1 will be described.

Part 1. Identifying shipping vessels related to the Dutch economy

The Dutch national accounts are mainly compiled with monetary data at business unit's level. For the sector Water transport (NACE H.50) only monetary data of the business units (the shipping companies) is available. There is no information on the shipping vessels that have generated the monetary values, through shipping activities, for these shipping companies. Identifying these shipping vessels will enable us to trace them using AIS data. With AIS data we will attempt to calculate the distance travelled by the shipping vessels and thereby measure the energy used and air emissions emitted. Therefore, first a method has to be developed to link shipping vessels to the Dutch shipping companies from the national accounts.

Information Sources

The primary source of information in terms of Shipping vessels related to the Netherlands is the annual inventory list of ships, provided by the Dutch department of Information and Transport. This list of ships was combined with the General Business Register and the Production Statistics.

The information sources are:

1. List with "Dutch" ships

   This list is received from the Dutch Ministry of Infrastructure and Water Management[3]. It contains all ships and vessels that are linked to the Netherlands based on one of the following 5 criteria:

   - Registered Owner
   - Ship Manager

[3] See Annex 2: Variables Shipping List for a list of all variables in the file.

- Operator
- Group Beneficial Owner
- [Ship] Flag Decode

Available data: 2011 to 2017

2. ABR (General Business Register)

All companies in the Netherlands contributing to the domestic product are in principle included in the ABR (General Business Register) of Statistics Netherlands The ABR serves as a sampling framework for surveys of companies for Statistics Netherlands. The central nature of the registration makes it possible to determine the various statistics in such a way that the outcomes of these statistics show a high degree of coherence, because they connect to each other or are related. A number of characteristics are available for each company, such as the economic activity according to the Standard Business Classification (SBI) and the company size based on the number of people employed. To be able to approach companies, the name and address are also registered.

Available data: 1994-2018

3. Production Statistics

Every year, Statistics Netherlands publishes data on the revenues, costs and assets of businesses in the Netherlands. The data contains financial information and data about the number of employees and on monetary value of fuel usage.

Available data: 2011 to 2018

Concepts and methodology

Air Emission Accounts (AEA [7])

Air emissions accounts (AEA) are one of six modules of Regulation (EU) No. 691/2011 on European environmental economic accounts. AEA contribute directly to the Union's policy priorities on climate change, green growth and resource productivity by providing important information on air emissions in a way that is compatible with the international UN system of national accounts (SNA) and its European version (ESA). Air emissions accounts (AEA) record the flows of gaseous and particulate materials from the national economy into the atmosphere. After joining the atmosphere, the emitted substances are out of any human control and become part of natural materials cycles and may induce several types of environmental impacts. AEA present air emissions in a breakdown by emitting economic activity; the latter comprise production and consumption activities. Air emissions comprise emissions of greenhouse gases as well as emission of air pollutants such as SO2, NOX, PM10, etc.

(Source: Eurostat Manual for air emissions accounts, 2015 edition)

Physical Energy flow Accounts (PEFA [6])

Energy accounts is a module of Regulation (EU) No. 691/2011 on European environmental economic accounts. It contributes directly to the Union's policy priorities on green growth and resource productivity by providing important information and indicators on energy in a NACE breakdown. PEFA present data on the physical flows of energy expressed in terrajoules in a way that is fully compatible with the concepts, principles, and data reported under the International System of National Accounts

(SNA) and European System of National and Regional Accounts (ESA). PEFA record energy flow data in relation to the economic activities of resident units of national economies. They present the supply and use of natural energy inputs, energy products and energy residuals. Economic activities comprise production, consumption, and accumulation.

(Source: Eurostat manual, *Physical Energy Flow Accounts (PEFA), Manual 2014 draft)*

Shipping vessels characteristics

In terms of nationality, shipping vessels have different characteristics. The main characteristics comprise:

1. Registered Owner

   The legal title of ownership of the vessel that appears on the ship's registration documents. It may be an Owner/Manager or a wholly-owned subsidiary in a larger shipping group; or a bank or one-ship company vehicle set up by the bank; or a "brass-plate" company created on paper to legally own a ship and possibly to limit liability for the "real" owners and/or benefit from off-shore tax laws. It may anyway be a legal-requirement of the flag-state with whom the ship is registered for the legal owner to be a company registered in that country (source: Qi [8]).

2. Commercial Manager

   The company designated by the ship owner or charterer to be responsible for the day to day commercial running of the ship and the best contact for the ship regarding commercial matters. (source: Qi [8])

3. Commercial Operator

   The company responsible for the commercial decisions concerning the employment of a ship and therefore who decides how and where that asset is employed. The direct beneficiary of the profits from the operations of the ship, this company may also be responsible for purchasing decisions on bunkers and port services. (source: Qi [8])

4. Group Beneficial Owner

   This is the parent company of the Registered Owner. It is the controlling interest behind its fleet and the ultimate beneficiary from the ownership. A Group Beneficial Owner may or may not directly own ships itself as a Registered Owner. (source: Qi [8])

5. Flag carrier

   The flag state of a merchant vessel is the jurisdiction under whose laws the vessel is registered or licensed, and is deemed the nationality of the vessel. A merchant vessel must be registered and can only be registered in one jurisdiction, but may change the register in which it is registered. The flag state has the authority and responsibility to enforce regulations over vessels registered under its flag, including those relating to inspection, certification, and issuance of safety and pollution prevention documents. As a ship operates under the laws of its flag state, these laws are applicable if the ship is involved in an admiralty case. The term "flag of convenience" describes the business practice of registering a merchant ship in a state other than that of the ship's owners, and flying that state's civil ensign on the ship. Ships may be registered under flags

of convenience to reduce operating costs, to avoid the regulations of or avo id inspection and scrutiny by the owner's country. Normally the nationality (flag) of the ship determines the taxing jurisdiction. (Source: 'Flag State', 2019 [10])

Dutch economy international shipping

The value added as generated by maritime shipping and reported by companies register in the following NACE REV2 groups:

H50	Water transport
H50.1	Sea and coastal passenger water transport
H50.1.0	Sea and coastal passenger water transport
H50.2	Sea and coastal freight water transport
H50.2.0	Sea and coastal freight water transport

Methodology

According to the European System of National and Regional Accounts (ESA 2010), the characteristic that represents the connection of a shipping vessel with a national economy is the owner and/or operator resident in the country:

"Fishing boats, other ships, floating platforms and aircraft are treated in the ESA as mobile equipment, whether owned and/or operated by units resident in the country, or owned by non-residents and operated by resident units. Transactions involving the ownership (gross fixed capital formation) and use (renting, insurance, etc.) of mobile equipment are attributed to the economy of the country of which the owner and/or operator respectively are residents. In cases of financial leasing, a change of ownership is assumed."

(Source: ESA 2010, Chapter 2. Units and grouping of units, 2.05)

Besides the ESA definition, it can be argued that contracts for shipping services are mostly made by operators and this group receives first hand benefits from the commercial activities of ships and vessels. Therefore the assumption can be made that operator resident in the country is the most probable characteristic relevant for the ships and vessels link with a national economy. This assumption of operator being the main characteristic was tested with data from the production statistics.

All companies that have reported in the "Jaarstatistiek Zeevaartbedrijven" (translation: annual statistics on sea shipping companies) of the Production Statistics (PS) were be investigated in the inventory list and their ships were traced in the inventory list of ships. As these companies have reported production and value added as part of the National Gross Domestic Product, their ships have most likely generated the value added for these companies. This was verified by analysing the turnover and monetary value of fuel usage in the reported variables of the company in the PS.

As a result it would show whether operator or a different characteristic such as flag, or a combination of characteristics can be identified as the most likely representation of shipping vessels having an economical link with a country. This method results in relevant ships and vessels to be identified for the AIS-data calculation in part 3.

Results

A ship sailing under the Dutch flag is not representative of companies linked to the Dutch economy. The top three companies with highest value added from shipping services, in 2017 of the NACE H.50 water transport group, do not have one ship sailing under Dutch flag of its total of 62 marine ships. These top three companies in highest value added have reported high fuel uses of crude oil or diesel, verifying that the productions are truly generated by shipping services.

Tracing these companies with highest value added in the ships inventory list, shows that the companies have no owner link with the Netherlands, neither as registered owner or beneficial owner. However the companies do have a commercial operator link, with the Netherlands as country of residence.

This PS and ships lists analysis shows that for the Netherlands, the operator with the Netherlands as residents' country is a probable representation of a ships economical link to a country. With this we used the operator characteristic to look up the shipping vessels connected to the Dutch economy, with the sources combined as follows:

The ABR and Ships list are connected by name of the operator. We used R with the text mining package StringDist for matching strings. The results are:

Year	Total number of companies in General Business Register (ABR)	Total number of ships	Number of ships matched	Number of ships without a match
2014	888	1930	610	1320
2015	899	2057	604	1453
2016	890	2059	570	1489
2017	820	2072	483	1589

Table 3 Results of matching ships to shipping companies

Next steps

As shown, more than 60% of the ships are not matched, so more research is needed to complete the first part of the process. In parallel, part two and three of the process can be developed. Part two is to retrieve and implement worldwide AIS data from Dutch shipping vessels to calculate travel distances. To start with, this will be done using AIS data available from the UN Global Platform (see below). The advantage is that the data is directly available and more importantly worldwide data. Part three is to compile energy used (for the PEFA) and air emissions (for the AEA) with the travel distance from AIS data of residents shipping vessels. We will describe more on this in the next deliverable.

3. Preliminary results on additional research

Besides the statistical products to be delivered, additional more experimental work was also performed. Described here is a first step into the investigation into indicators of economic trend changes. Unfortunately, no work has been done on the reference frame of ships and on the port to port distance matrix. Hopefully, work on this can be done in the next couple of months.

Indicators of economic trend changes

The Netherlands has an open economy, in which transport in general and shipping in particular plays a very important role. The expectation is therefore that important trends or changes in trends of the Dutch economy are strongly correlated with developments in activity in maritime logistics. There is a particular interest in developing early indicators for important changes in trends or turning points in economic activity.

For application in financial markets, methods exist that rely on patterns in the topology of networks (Squartini et al., 2013, [12]) and on the spreading of information to build early indicators (Gatfaoui & de Peretti, 2019, [10]) with some success.

The purpose of this brief note is to outline a way to translate this method to the setting of (international) shipping by means of AIS, with the purpose of building an early indicator for trend changes in the real economy.

Method

In the papers of Squartini et al. (2013, [12]) and of Gatfaoui & de Peretti (2019, [10]) banks and financial markets (stock markets) respectively are modelled as a complex (networked) dynamical system. In these models either the banks or the individual stock exchanges are the nodes in such a network. For banks it is mutual debt ('exposure') is monitored as a function of time, whereas for stock markets the price index is the quantity that is monitored as a function of time. Time-series models are used to assess to what extent there is an information flow between any two nodes (hence a causal link) in this network, from which an asymmetric adjacency matrix can be constructed. This can be done over finite time windows, moving over the entire available time series, so that changes in this flow of information can be measured.

Using the asymmetric adjacency matrices, triangular clusters or 'motifs' of information transmission patterns over time windows are isolated, and their evolution over time is analysed. In this context a 'motif' is a particular fixed pattern of connections between a small number of nodes, that occurs very frequently in the data. An example is for instance a star-motif: a single central node that has a connection with a number of other nodes, where those other nodes have no mutual connections. As a result, it is possible to characterise the way nodes communicate with their neighbours in the very short term. From this it is then possible to recover the mapping of information spreading across the network, and over time i.e. information spreading patterns in a dynamic network. There is strong empirical evidence that a sudden increase in correlations among all components of a financial network precedes a critical transition. Monitoring the correlation therefore opens up the possibility of an early indicator for such transitions.

In the context of shipping it is the port, where freight is loaded or offloaded, that are the nodes in a network. Since goods are physically transported between ports there is already directional information in the available data, contrary to what is the case in the study of financial markets. In principle this means that dynamic and asymmetric adjacency matrices can be constructed directly from the AIS data. The links, or adjacency, in this case are the individual legs of every shipping journey: pairs of harbours that are visited successively by a cargo vessel are linked.

Some experimentation with the length of time window is necessary. It is possible that with time windows that are quite short, routes between ports that are less frequently travelled because of lower volumes of cargo, will disappear from the network, so that only major nodes will remain. Very long time windows may have the disadvantage of creating very many potentially correlated motifs, out of which only a few have a major influence on the dynamics of the network. Also, the dynamical character of the network is suppressed by excessive averaging. If there is no apparent change in the adjacency matrices at all, then all the edges in the network appear to be completely fixed and this method is no longer informative.

Operationalisation using AIS data

The direct translation of the methods of Squartini et al. (2013, [12]), or Gatfaoui & de Peretti (2019, [10]) implies using the information on origins and destinations of ships to determine the edges in the network, and follow rapid changes in the network topology over time. However there is additional information that can be extracted from AIS data, which is the residence time of ships in ports. For larger types of ships, and/or volumes of cargo that their operating logistics companies handle, the in-port residence time is likely to be minimised. Unless there is a need for repairs for such ships the subsequent route(s) may already be scheduled while a ship is still en route between any two ports. Such optimised waiting times might therefore not be very informative for the purposes of measuring varying economic activity. For smaller ships or contractors, the local supply and demand for transport may be a strong determinant of the residence time in the port as well as the next destination of any given ship.

It is worthwhile to stratify the population of ships and apply the same method on each of the separate strata, and in addition use in-port residence times as an auxiliary indicator of economic activity.

While for the purposes of the Dutch economy, the Dutch ports are most of interest, these ports are embedded in the international network. While it is not necessary, and probably not feasible, to include every port in the world in the network to be modelled, it is likely desirable to include most or all of the (cargo) ports in North-western Europe, as well as some major global destinations.

4. Quality of AIS data from EMSA

Within this WP, we use European terrestrial AIS data from the European Maritime Safety Agency (EMSA) of the period January 2016 until December 2017 for our investigations. EMSA receives the data directly from the member states' coastguards. As EMSA's data could provide a common source for European NSI's, we performed an analysis to assess the quality of EMSA data compared to national AIS data. Therefore, national AIS data from Greece and Poland is compared to AIS data from EMSA. Note that we did not perform this analysis for the Dutch AIS data for inland waterways (IWW), as each NSI would have to use their national AIS IWW data. Also, we did not perform a separate quality assessment for the UN Global Platform as this is not our own source and a more basic quality assessment was already performed by ONS.

When investigating the quality of AIS data it is important to keep in mind that:

- AIS is a radio signal, parts of the messages can get lost or scrambled due to factors such as meteorology or magnetics.
- Messages are transmitted encoded. If a message is distorted in a small part only of the encoded message, for example due to meteorological influences, this can result in a larger erroneous part in the decoded message. That is, an error in one transmitted 'byte' can result in an error in one or multiple fields in the decrypted message. Most of the times, these errors are detectable as the result yields an invalid variable, but sometimes they result in valid variables. For instance, coincidentally the resulting MMSI (Maritime Mobile Service Identity) can be a technically valid, but incorrect MMSI, resulting from an erroneous detection. These errors can arise for every variable, so this can for example result in erroneous latitude and longitude, yielding faulty locations that are quite far away from the actual location of the ship. In turn, this can result in a very high journey distance of ship.
- Receivers have timeslots in which data is received. In busy areas with many ships, not all data from all ships may fit into this time slot. Resulting in the loss of data on some ships in that time slot.
- Ships can turn off their AIS transponder resulting in the disappearance of a ship.
- AIS was intended originally for safety at sea, to warn nearby ships. As it was not meant for producing statistics, the variables entered manually by the shippers are not always reliable.
- We only have terrestrial AIS EMSA data. Terrestrial AIS receivers only pick up signals within the range of about 40 sea miles. Therefore, land receivers have a very limited coverage of signals transmitted from sea which results in loss of information of ships on open sea.

This chapter describes the results on the further investigation of the quality of the AIS data from EMSA. First, we describe the quality and metadata framework we have chosen. This chapter ends with the comparison of the densities and frequencies of the European AIS data and the national AIS data of Denmark, Greece and Poland.

Quality framework

To provide an overview of different aspects of the quality of the AIS data from EMSA, we used a preliminary framework for national statistical offices to conceptualise the quality of big data. This Big

Data Quality framework is a deliverable of the Big Data Quality Task Team from the UNECE/HLG project, The Role of Big Data in the Modernisation of Statistical Production [13].

This preliminary framework was developed building on dimensions and concepts from existing statistical data quality frameworks. It provides a structured view of quality at three phases of the business process:

- Input – acquisition, or pre-acquisition analysis of the data;
- Throughput –transformation, manipulation and analysis of the data;
- Output – the reporting of quality with statistical outputs derived from big data sources.

In the table beneath the big data quality framework for the input phase of the business process is reported. This is done only for the European AIS data source of EMSA. In the table we filled out the factors to consider for each quality dimension. See annex 3 for a more detailed rating of this quality framework. In this annex we rated different indicators for each quality dimension.

Hyper dimension	**Quality Dimension**	**Factors to Consider**	**Remarks considering source**	**Conclusion**
Source	*Institutional/ Business Environment*	Sustainability of the entity-data provider	Data is transmitted through the ether. Many data providers available, who deliver the same data.	OK
		Reliability status	Highly reliable	OK
		Transparency and interpretability	Used filters by the data provider are not always clear. However, the data is standardized, which means there is not much room for interpretations	OK
	Privacy and Security	Legislation	Especially on the inland waterways, there are some cases where owners of the vessels have privacy issues. EMSA data does not contain inland AIS.	OK
		Data Keeper vs. Data provider	Data Keeper are the member states, EMSA is Data provider. However, due to legislation, the owner of the data should be the owner of the vessel	OK
		Restrictions	Some ships are owned by a private person, which puts some restrictions on the aggregation level of the data	OK
		Perception	From the owners of the ships, the use of the data might be perceived negative. Using a cautious publication level should minimize negativity	OK
Metadata	*Complexity*	Technical constraints	Data is stored in the NMEA data format. To decode the messages, libraries can be used that are available for different programming languages	OK
		Whether structured or unstructured	The data is highly structured	OK
		Readability	Data is not human readable, although well readable by machines.	OK

		Presence of hierarchies and nesting	No nesting available, but different record types	OK
	Completeness	Whether the metadata is available, interpretable and complete	Available, but one has to search the internet for it. Elements that are not processed automatically have to be put in by owner of the ship, who might not always be aware of all the details	OK
	Usability	Resources required to import and analyse	When dealing with Europe wide data for a longer time span, big data infrastructure and skills are needed	OK
		Risk analysis	Not applicable. These skills and infrastructure are also necessary for other sources	OK
	Time-related factors	Timeliness	The data is high velocity. High quality data can be available in an instance	OK
		Periodicity	Collection time and reference period is about the same.	OK
		Changes through time	Not applicable. The data is collected and stored as is.	OK
	Linkability	Presence and quality of linking variables	Due to transmission errors, the IMO number necessary to link to maritime statistics are sometimes wrongly received. We developed a method to deal with this.	OK
		Linking level	Linking can be at vessel level based on the IMO/MMSI number and precise geographical locations	OK
	Coherence - consistency	Standardisation	standardized	OK
		Metadata available for key variables (classification variables, construct being measured)	yes	OK
	Validity	Transparency of methods and processes	The data EMSA receives from the member states is down-sampled using a 6 minute period. More information is needed.	NOT OK
		Soundness of methods and processes	We do not know if there are critical flaws in the processes of filtering the data that would limit potential use of the data.	NOT OK?
Data	*Accuracy and selectivity*	Total survey error approach	See deliverable 4.3 [5] and 4.2 [4]	OK
		Reference datasets	See deliverable 4.2 [4]	OK
		Selectivity	All ships are present in the data. Spatial coverage seems sufficient.	OK
	Linkability	Quality of linking variables	Linking is done based on IMO/MMSI numbers for vessels and Lat/Lon for ports	OK
	Coherence - consistency	Standardized concepts	All key variables are standardized (IMO's, Lat/Lon)	OK
		Coherence with metadata	yes	OK

	Validity	Transparency of methods and processes	Not transparent (see also metadata validity)	NOT OK
		Soundness of methods	Not transparent (see also metadata validity)	NOT OK?

Table 4: quality framework factors to consider

This quality framework confirms that the quality of the AIS data from EMSA is good. Table 2.1.1 shows that almost all factors of the quality framework are judged as mostly positive. Due to transmission errors, the IMO-numbers necessary to link AIS data to maritime statistics are sometimes wrongly received. We already developed a method to deal with this (see previous deliverable [4]). We only judged the spatial coverage, transparency and soundness of methods and processes for the metadata and the data as not ok because not all European coastal areas are covered and DZ pre-processes the data, but the documentation on how they do this is not available to us.

5. Infrastructure platforms

Big Data Test Infrastructure

BDTI is a cloud infrastructure provided by the European Commission's Connecting European Facility (CEF). From a technical point of view, this platform is based on Amazon Web Services (AWS), that offers virtual environments allowing public organizations to:

- experiment with big data services, methods and tools;
- launch pilot projects (e.g. AIS pilot for our case-study).

The environment, technical parameters and tools were established during a technical seminar in April 2019 at CBS in Heerlen. The meeting was attended by members of this Work Package (ESSnet WPE), Eurostat and representatives of the consulting companies KPMG and Deloitte, who supported WPE in technical aspects of the BDTI. We had access to BDTI from 5 May to 2 December 2019. During that period there was a technical break (five weeks) during a period of inactivity, so that the period could be used as an extension.

Architecture overview

Figure 19 illustrates the architecture of the implemented environment.

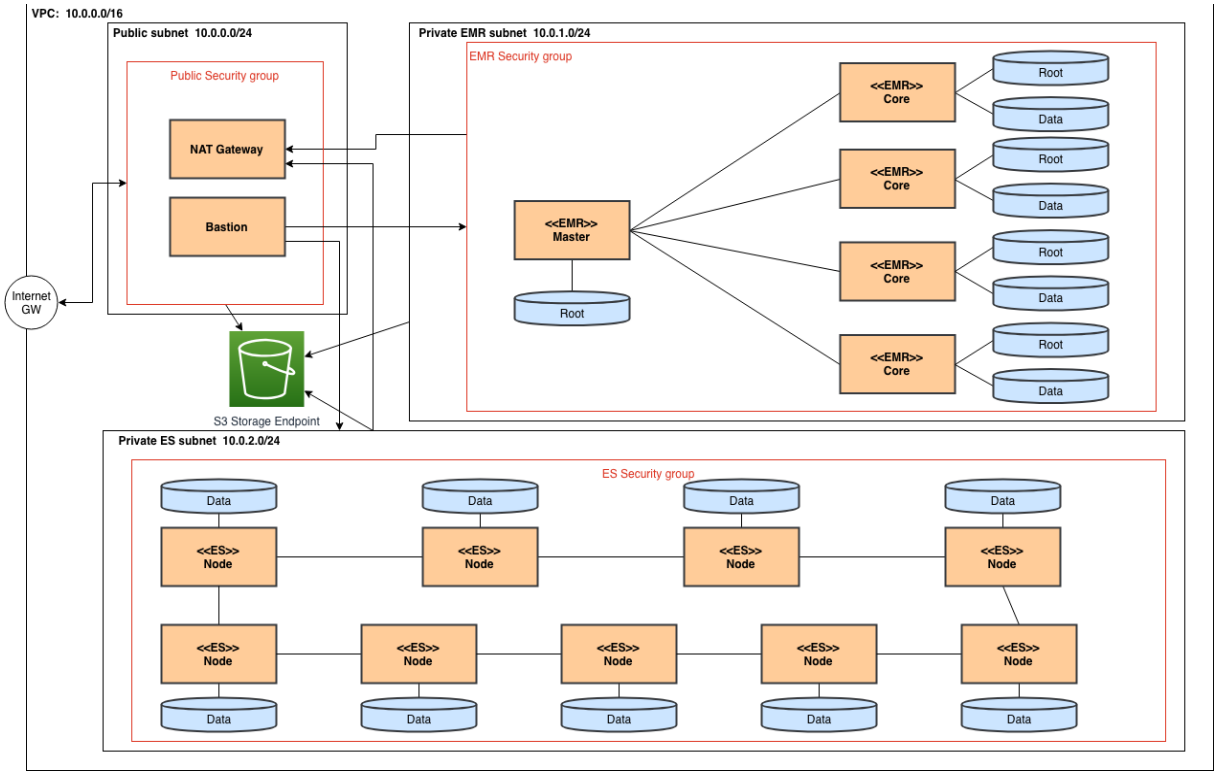


Figure 19 Architecture BDTI, source: Document ais-documentation.md provided by Brecht Verhoeve - Deloitte

The BDTI environment was hosted in the Virtual Private Cloud (VPC) with CIDR range 10.0.0.0/16. Three subnetworks were deployed in this VPC: a public sub-network (public subnet) and two private sub-networks. In the public subnet, one instance was deployed, the bastion host acted as a single-point of access for all other instances.

The private EMR subnet was deployed as a computing cluster providing applications like the Hadoop, Spark, Scala, Python, R, Zeppelin, R, RStudio Server and Shiny Server. This cluster consisted of 1 master and 4 core nodes.

The private ES subnet was deployed Elastic Stack providing Elasticsearch and Kibana. This cluster consisted of 9 nodes.

Experiences so far

For our work, we uploaded decoded AIS data from EMSA (2016-2017) to the Hadoop File System (HDFS) and made them available to all project members. We also had an additional data storage space, that is the S3 bucket, which is specific for AWS solutions. In our case, S3 played the role of a data storage buffer before the data finally reached HDFS.

One of the advantages of the BDTI environment and a valuable experience was the opportunity to use various tools for processing and analysing AIS data. The most important of them include:

- Hadoop Hue – an open-source user interface for Hadoop components, allowing the user to access HDFS files via web browser and interact with them.

  As in the previous project, we had two types of data:

  - Static (Messages)

  - Dynamic (Locations)

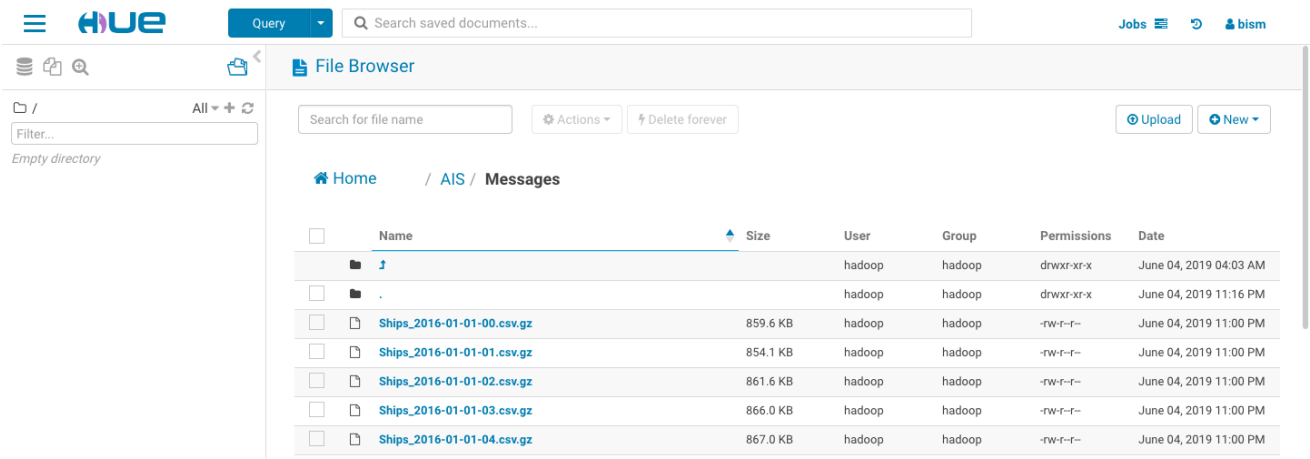


Figure 20 Hue – BDTI, source: screenshot of Hue Hadoop for BDTI - own source

- Apache Spark – an open-source distributed general-purpose cluster-computing framework. Spark provides an interface for programming entire clusters with implicit data parallelism and fault tolerance.

Access to AIS data is possible from Spark in two ways:

- spark-shell – an interactive shell for the API through Spark's.
- spark-submit – to launch applications on a cluster.

In the first case, on the command line:

```
> sc

>val rawdata = sc.textFile ("hdfs://ip-10-0-1-66.eu-west-1.compute.internal:8020/AIS/Locations/Locs_2016-01-01-00.csv.gz");

>val splitdata = rawdata.map(_.split(","));

>val data = splitdata.filter(x => x(0)!="mmsi");

>data.count();
```

Result:

```
scala> val rawdata = sc.textFile ("hdfs://ip-10-0-1-66.eu-west-1.compute.internal:8020/AIS/Locations/Locs_2016-01-01-00.csv.gz");
rawdata: org.apache.spark.rdd.RDD[String] = hdfs://ip-10-0-1-66.eu-west-1.compute.internal:8020/AIS/Locations/Locs_2016-01-01-00.csv.gz MapPartitionsRDD[5] at textFile at <console>:24

scala> val splitdata = rawdata.map(_.split(","));
splitdata: org.apache.spark.rdd.RDD[Array[String]] = MapPartitionsRDD[6] at map at <console>:25

scala> val data = splitdata.filter(x => x(0)!="mmsi");
data: org.apache.spark.rdd.RDD[Array[String]] = MapPartitionsRDD[7] at filter at <console>:25

scala> data.count();
res2: Long = 58149

scala>
```

In the second case, on the command line:

```
Spark-submit --class AIStest hdfs://ip-10-0-1-66.eu-west-1.compute.internal:8020/user/bism/test/ais_aws_test_2.11-0.1.jar
AIS/Locations/Locs_2016-01-01-00.csv.gz
```

Result:

```
19/09/29 20:20:58 INFO SparkContext: Starting job: count at first.scala:19
19/09/29 20:20:58 INFO DAGScheduler: Got job 0 (count at first.scala:19) with 1 output partitions
19/09/29 20:20:58 INFO DAGScheduler: Final stage: ResultStage 0 (count at first.scala:19)
19/09/29 20:20:58 INFO DAGScheduler: Parents of final stage: List()
19/09/29 20:20:58 INFO DAGScheduler: Missing parents: List()
19/09/29 20:20:58 INFO DAGScheduler: Submitting ResultStage 0 (MapPartitionsRDD[3] at filter at first.scala:18), which has no missing parents
19/09/29 20:20:58 INFO MemoryStore: Block broadcast_1 stored as values in memory (estimated size 3.7 KB, free 413.7 MB)
19/09/29 20:20:58 INFO MemoryStore: Block broadcast_1_piece0 stored as bytes in memory (estimated size 2.2 KB, free 413.7 MB)
19/09/29 20:20:58 INFO BlockManagerInfo: Added broadcast_1_piece0 in memory on ip-10-0-1-66.eu-west-1.compute.internal:44703 (size: 2.2 KB, free: 413.9 MB)
19/09/29 20:20:58 INFO SparkContext: Created broadcast 1 from broadcast at DAGScheduler.scala:1201
19/09/29 20:20:58 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 0 (MapPartitionsRDD[3] at filter at first.scala:18) (first 15 tasks are for partitions Vector(0))
19/09/29 20:20:58 INFO YarnScheduler: Adding task set 0.0 with 1 tasks
19/09/29 20:20:59 INFO ExecutorAllocationManager: Requesting 1 new executor because tasks are backlogged (new desired total will be 1)
19/09/29 20:21:03 INFO YarnSchedulerBackend$YarnDriverEndpoint: Registered executor NettyRpcEndpointRef(spark-client://Executor) (10.0.1.158:45720) with ID 1
19/09/29 20:21:04 INFO ExecutorAllocationManager: New executor 1 has registered (new total is 1)
19/09/29 20:21:04 INFO TaskSetManager: Starting task 0.0 in stage 0.0 (TID 0, ip-10-0-1-158.eu-west-1.compute.internal, executor 1, partition 0, RACK_LOCAL, 7962 bytes)
19/09/29 20:21:04 INFO BlockManagerMasterEndpoint: Registering block manager ip-10-0-1-158.eu-west-1.compute.internal:40043 with 3.8 GB RAM, BlockManagerId(1, ip-10-0-1-158.eu-west-1.compute.internal, 400
43, None)
19/09/29 20:21:05 INFO BlockManagerInfo: Added broadcast_1_piece0 in memory on ip-10-0-1-158.eu-west-1.compute.internal:40043 (size: 2.2 KB, free: 3.8 GB)
19/09/29 20:21:05 INFO BlockManagerInfo: Added broadcast_0_piece0 in memory on ip-10-0-1-158.eu-west-1.compute.internal:40043 (size: 24.0 KB, free: 3.8 GB)
19/09/29 20:21:05 INFO TaskSetManager: Finished task 0.0 in stage 0.0 (TID 0) in 1968 ms on ip-10-0-1-158.eu-west-1.compute.internal (executor 1) (1/1)
19/09/29 20:21:05 INFO YarnScheduler: Removed TaskSet 0.0, whose tasks have all completed, from pool
19/09/29 20:21:05 INFO DAGScheduler: ResultStage 0 (count at first.scala:19) finished in 7,675 s
19/09/29 20:21:05 INFO DAGScheduler: Job 0 finished: count at first.scala:19, took 7,751774 s
Lines in dataset: 58149
19/09/29 20:21:05 INFO SparkUI: Stopped Spark web UI at http://ip-10-0-1-66.eu-west-1.compute.internal:4040
19/09/29 20:21:06 INFO YarnClientSchedulerBackend: Interrupting monitor thread
19/09/29 20:21:06 INFO YarnClientSchedulerBackend: Shutting down all executors
19/09/29 20:21:06 INFO YarnSchedulerBackend$YarnDriverEndpoint: Asking each executor to shut down
19/09/29 20:21:06 INFO SchedulerExtensionServices: Stopping SchedulerExtensionServices
(serviceOption=None,
 services=List(),
 started=false)
19/09/29 20:21:06 INFO YarnClientSchedulerBackend: Stopped
19/09/29 20:21:06 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
19/09/29 20:21:06 INFO MemoryStore: MemoryStore cleared
19/09/29 20:21:06 INFO BlockManager: BlockManager stopped
19/09/29 20:21:06 INFO BlockManagerMaster: BlockManagerMaster stopped
19/09/29 20:21:06 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
19/09/29 20:21:06 INFO SparkContext: Successfully stopped SparkContext
19/09/29 20:21:06 INFO ShutdownHookManager: Shutdown hook called
19/09/29 20:21:06 INFO ShutdownHookManager: Deleting directory /mnt/tmp/spark-5f2a9b69-a649-42fe-9dee-684fb8b93d29
19/09/29 20:21:06 INFO ShutdownHookManager: Deleting directory /mnt/tmp/spark-63351a79-771f-46bc-a63b-6b36583e5f59
[hadoop@ip-10-0-1-66 ~]$
```

Generally, we used code developed in Scala/Python and processed by Apache Spark.

- Apache Zeppelin – a web notebook that enables interactive data analytics. Spark is also supported by Zeppelin. Apache Spark is supported in Zeppelin with Spark interpreter group which consists of five interpreters. In our cases we use %spark – Spark Interpreter which creates a Spark Context and provides a Scala environment.

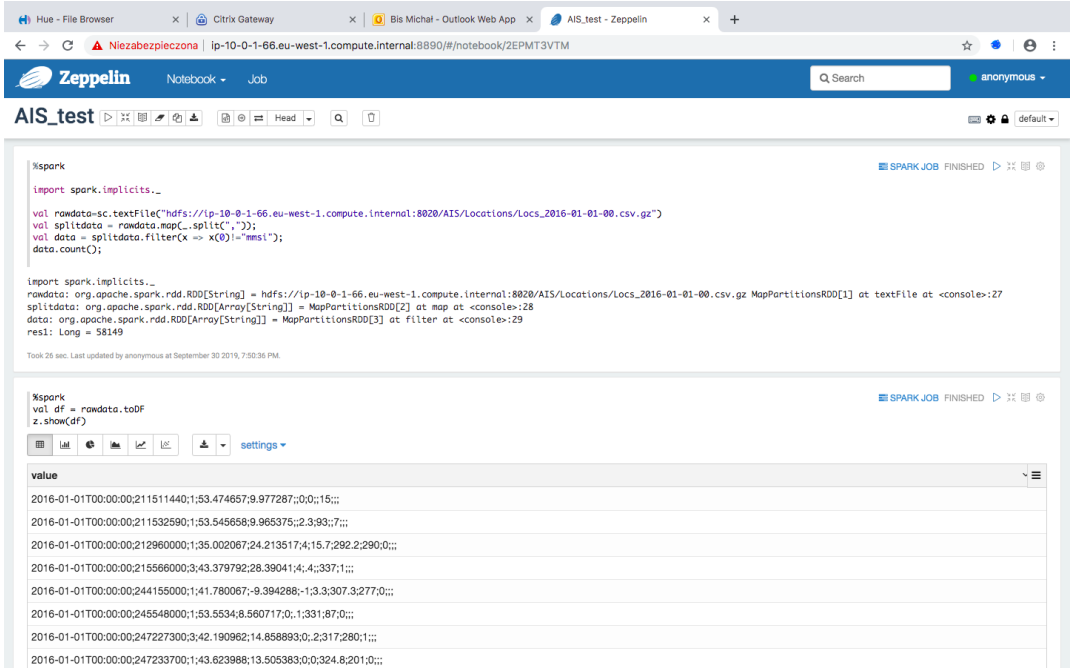


Figure 21 Zeppelin – BDTI, source: screenshot of Apache Zeppelin for BDTI - own source

- Jupyter Notebook – an open-source web application allowing to create and share documents containing live code, equations, visualizations and narrative text. It can also be used for Scala development with the Spylon-kernel.

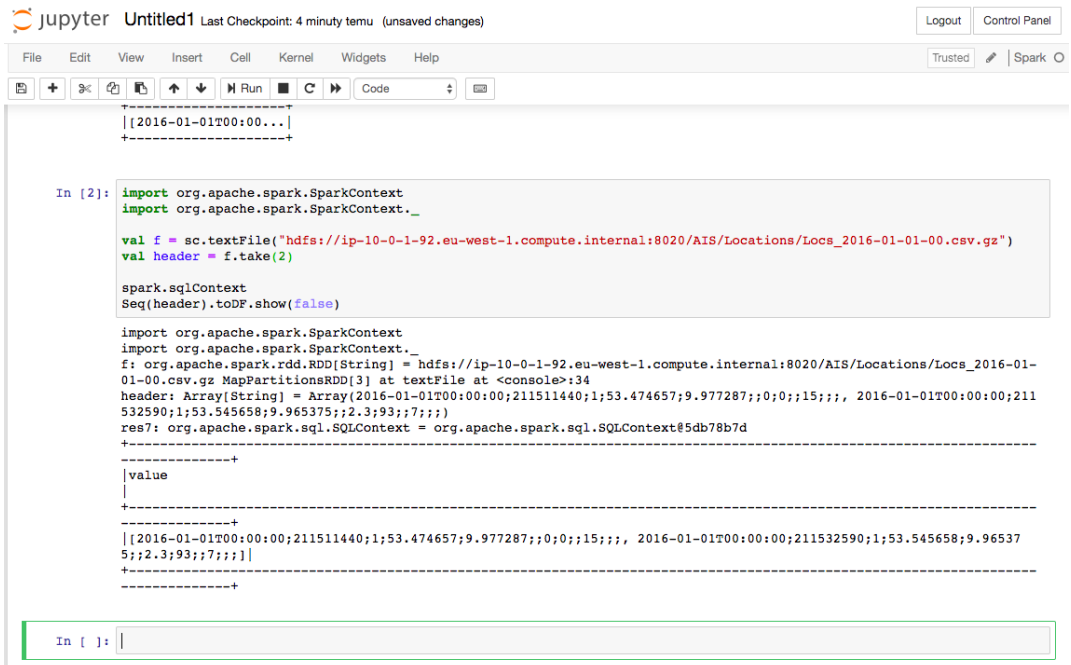


Figure 22 Jupyter Notebook – BDTI, source: screenshot of Jupyter Notebook for BDTI - own source

Lessons learned

BDTI was a common work environment for our package members. The advantage of such a cloud solution was the opportunity to exchange experience, work together and increase knowledge in the context of Big Data tools. BDTI provided high computing power, which is essential when processing AIS data.

Limitations

The main BDTI limitation was a relatively short availability-time of the environment for ESSnet members (May – December 2019). Full configuration, testing of the BDTI environment and the implementation of all the necessary tools, as well as the need to archive the results of work, data (due to the closure of the environment) seems disproportionate in the context of the lifetime of the environment for the needs of the WPE team.

Recommendations for future use

From a technical point of view, the BDTI cloud environment was fully adapted to the needs of ESSnet members. The quality of the service provided, in particular its reliability and performance, was at a very high level. The procedure for reporting problems related to BDTI services and technical support (Deloitte) is also assessed positively.

BDTI is a great environment to experience the technology and all modern tools necessary to design a complete statistics production process, based on the AIS data.

UN Global Platform

The Global Working Group (GWG) on Big Data for Official Statistics was created in 2014, as an outcome of the 45th meeting of the UN Statistical Commission. The UN GWG provides strategic vision, direction and coordination of a global programme on Big Data for official statistics, including for indicators of the 2030 agenda for sustainable development. It also promotes practical use of Big Data sources, while promoting capability building, training and sharing of experiences. Finally, the UN GWG fosters communication and advocacy of use of Big Data for policy applications and offers advice in building public trust in the use of Big Data from the private sector. To achieve these goals, various task teams were formed including Big Data and the SDGs, Mobile Phone Data, Satellite Imagery and Geo-Spatial Data and Scanner Data. The AIS Task Team was originally established during the United Nations Global Platform for Official Statistics workshop at Statistics Denmark. Similar to Eurostat's ESSnet, the main objective is to use AIS data for the creation of official statistics. In 2019, the Data Science Campus of UK ONS presented their work on faster indicators of economy activity for which AIS data was also used on the 5th Big Data Conference in Rwanda. After this, the AIS task team was reinvigorated to expand its original objectives and to include members from developing countries.

The AIS Task Team will develop algorithms and methodologies; conduct trainings – taking into account existing initiatives in this area – for using AIS data to create official statistics; and develop experimental data fit-for-purpose. AIS is applied in a diverse range of areas such as freight transport, traffic within ports, international trade/economic indicators, CO2 emission, fishery and rescue operations. The various areas, different organizations and countries are also expressed in the members of the task team. The members of this ESSnet's Working group also participate. The AIS Task Team will use the UN

Global Platform for the global collaboration on developing, and publishing of the algorithms, methods and training.

The UN Global Platform (UNGP)

The UN Global Platform is created for:

- Exchanging ideas and methods for processing, analysing and visualizing big data

  The task team organises monthly meetings. All members can attend, discussing for example work in progress and future plans. UN also provides GitLab and a wiki containing all information from the task team.

- Storing big data sources and enable processing, analysing and visualizing the data

  To analyse AIS data, two components are available at the UN Global Platform:

  - Stealth: a browser-based dynamic spatial visualization tool that can represent many layers of geospatial data along with their relationships
  - Jupyterhub: a console-based, interactive web application that streamlines developing, documenting and executing code

  The computation is done through the location service on the platform, which uses Apache Spark / Hadoop distributed environment to enable processing in reasonable time. It also uses the GeoMesa open source library to facilitate large-scale geospatial querying and analytics of AIS data.

  Worldwide AIS data is made available stemming from two private providers: exactEarth and Orbcomm. In the summer of 2019, data quality from both providers was compared. On the basis of the number of messages and the number of unique MMSI-numbers, exactEarth was selected as the future provider.

- Demonstrating the value of Big Data in better decision-making through pilots and case studies

  Pilot cases and algorithms used are described. UK ONS started with port traffic and time spent in port.

- Providing training materials and workshops for capacity building

  In September 2019, an online training session was provided describing the AIS data sets, the use of GitLab, cleaning and preparation, work by UK ONS and future work.

  A handbook is being developed, which will be describing the AIS data sets, cleaning and preparation, case studies, the use of AIS data at the global platform and future work. ESSnet's members also provide input for this.

Experiences so far

Once provided access, there were some scripts available that could be adjusted. This enabled fast access to the data. The data that is available at the UN Global Platform is pre-cleaned by the service provider for advanced analytical techniques. However, gaps in the messages can still occur in the data. Communication about this can be done with the technical team via Slack or via e-mail. Compared to BDTI, this bigger structure of people providing help was useful. Working with Stealth was sometimes very slow, and the number of dimensions is very limited.

Results from the task force

As mentioned already, the UN Global Platform had provided workshops, access to data and a handbook. In addition, there will also be efforts to obtain a single database on ship characteristics, which is beneficial for a wide range of worldwide governmental organizations.

Lessons learned

It is very interesting to being able to compare different methods. For example, UK ONS worked on port traffic and time spend in port. This algorithm is comparable to the ESSnet's work on port visits. In addition, UK ONS studied the use of these statistics as fast economic indicators, where ESSnet will also focus on.

Limitations

The main limitation lies on the software and infrastructural level already mentioned. The advantage of this already being available, is also the disadvantage. The tools or infrastructure cannot be arranged according to specific needs. For example, it would be useful to have a visualization tool. Stealth, which is provided by the UN Global Platform, is very limited. Also, the wide range of backgrounds and interests could hinder focus, but overall it provides a large body of information and data.

Recommendations for future use

In the upcoming months we will try to develop an algorithm to determine distance travelled to use as a proxy for air emissions by Dutch ships. As worldwide data is needed for this, the data should provide an excellent source for this.

6. WPE architecture presented in BREAL

In this ESSnet, work package WPF is defining the Big data REference Architecture and Layers, abbreviated as BREAL. The aim of this architecture is to help the development of standard solutions and services to be shared within the ESS and beyond. Therefore, it is beneficial to apply BREAL to the AIS solutions created during this work package. Beneficial for us, so that we can see whether we can share solutions, services and components within our work package, but also with other work packages. And it is beneficial for work package WPF, so that they get feedback on how the architecture is used and whether and where improvements are needed.

Information Architecture

The BREAL information architecture consists of three layers: the raw data layer, the convergence layer, and the statistical layer. In Figure 23, the WPE information architecture is shown, with the three layers and the information objects.

Figure 23 WPE Information Architecture

Raw data layer

The raw data layer contains all the raw data needed to create the statistics required. In case of work package WPE, the most important raw data is the AIS (Automatic Identification System) data in its purest form, which are the messages sent between the AIS transponders and the AIS antennas (both land-based and satellites). This data is gathered by various parties, on national, European and worldwide level.

Note that as a statistical office, we do not get this purest level of raw data, because the parties involved, already pre-process the data. For example, the data on national level is more detailed than on European and worldwide level, because in the latter case, the data is already filtered and combined (grouped). This is not a problem, as long as one knows what type of pre-processing was applied. In the ideal situation, the statistical office gets the data as specified in the convergence layer. Here, the data

is defined on such a level that is it exactly right to produce all statistics required. And, preferably, all details only needed by the domain-experts in the field are no longer present (hide the complexity).

In order to produce the fishing fleet statistics and emission statistics, ideally, it is also required to receive weather data. You need to know whether and where the fishing fleet was able to fish, and how much energy was needed to travel due to water and wind conditions. Because at the time of writing, we did not know yet how to receive this data, nor how it would look like, we placed it is the raw data layer. If the data is provided as required, it is actually part of the convergence layer.

When comparing and discussing the information architecture with our colleagues from work package WPD, which is also about sensor data, namely smart (energy) meters in buildings, we discovered that we both need weather data and that we both did not have this data yet. This shows how the architectures can help and how it can result in combining our efforts.

Convergence layer

The convergence layer contains the data on such a level that it is sufficient to create all statistics required. The data is represented as units of interest. In case of AIS, our main focus are the ships (at least to be able to ID the ships) and the position and movements (direction, speed and depth) of these ships. For example, looking at the raw AIS data, consisting of 27 different type of messages, only message 5 is important to identify the ship, and messages 1, 2, 3 and 21 are important to derive the position and movement.

Besides the actual data, we also need some registers to enrich the data. In case of WPE, we need the business register, the ship register and a port register.

The ship register (for example, Lloyd's Register) is needed to get the ship details, like type of ship, type of engine, type of cargo, owning company, flag of ship. This information is needed to create correct statistics. For example, for the emission statistics the nationality of a ship determines whether the emission should be counted or not (thus, not whether it travelled in the national waters).

The port register is needed to know where the ports are (physical location) and what type of ports they are (what type of cargo is supported). Because, in not all countries, such a port register exists, it might be needed to derive this port register from the AIS data itself.

Statistical layer

The statistical layer, contains all concepts needed for the required statistics. Work package WPE actually consists of four subprojects and various products (concepts):

- The purpose of the port visits statistics is to create the mandatory F2 dataset, containing the port visits.
- For the inland waterway statistics in The Netherlands, the AIS data is used to resolve the under-coverage in the current statistics. For this, the journeys of ships is needed.
- For the emissions and energy used, the emissions of ships need to be derived.
- For the fishing fleet statistics, the traffic and activity (behaviour) of fishing ships is needed.

Of course, many more statistics and concepts might be needed in the (near) future. In the ideal case, we only need to derive a new concept in the statistical layer, using the units provided in the

convergence layer. This would imply that we have chosen the convergence layer at the right level. Of course, we will not be able to look into the future, so there will be cases where we have to extend the units in the convergence layer, or create new ones.

In a more formal notation, the WPE information architecture is as follows:

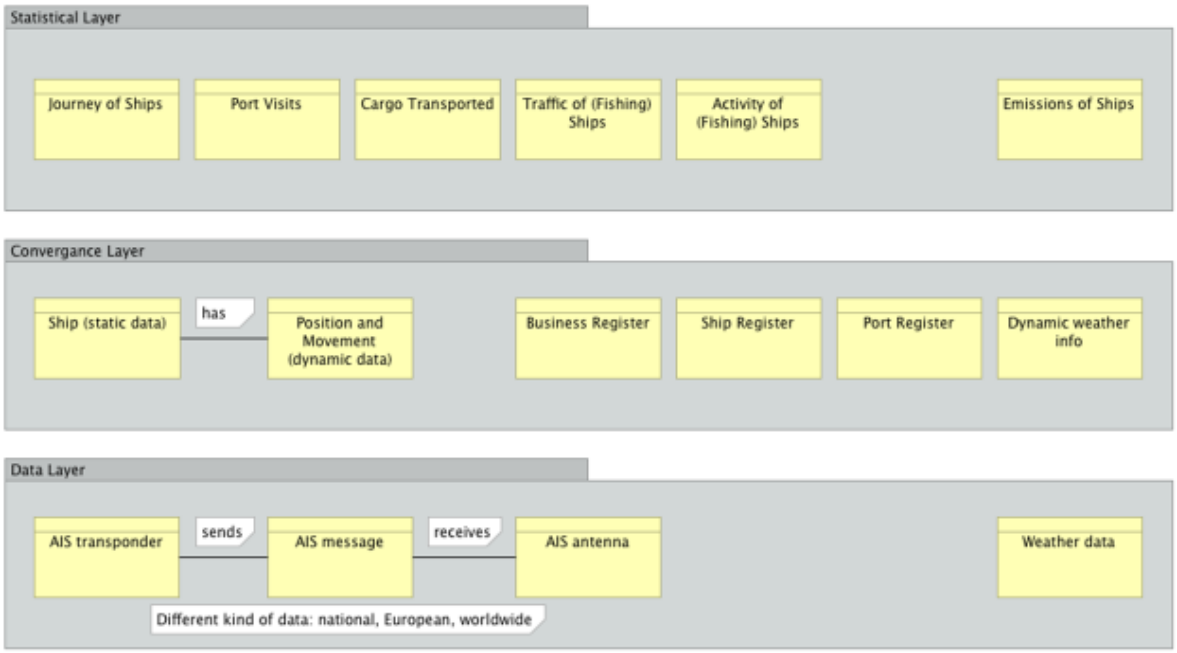


Figure 24 WPE Information Architecture, a more formal model

Application Architecture

The BREAL architecture has adopted the Archimate language for describing the different architecture and their relationships. The beauty of that language is that different colours a re used for the different types of architecture: yellow is used for the business architecture, blue for the application architecture, and green for the technical architecture.

For the AIS solution to resolve the under-coverage of the Dutch inland waterway statistics, the application architecture is shown below. The process consists of the following steps:

- Receive AIS data

  The first step in the process is to receive the AIS data. The national AIS data is provided by Rijkswaterstaat, which is part of the Dutch Ministry of Infrastructure and Water Management and responsible for the design, construction, management and maintenance of the main infrastructure facilities in the Netherlands. The data received is 22 GB per week and is currently provided via hard-disk. Of course, this will change in the near future. The file is copied from the hard-disk to the production environment of Statistics Netherlands. So, the application service is "Copy file" and the technical service is a regular "File copy".

- Decode and filter AIS data

  The next step is to decode and filter the AIS data. The data is in a specific technical format and there are special Python libraries, like libAIS, to decode the data in a more friendly machine-readable format. Raw AIS data is decoded and immediately filtered, resulting in two files: a static

file (record type 5), and a dynamic file (record types 1, 2, 3 and 21). All other records are ignored. Only the variables MMSI, speed and location are selected.

The application component "AIS data preparation" provides the application services "Decode and filter AIS data". In fact, to be more precise, there are actually three application services: decode AIS data, filter records, and filter variables.

- Split and combine per ship

  Now, only the required data has been selected and filtered, the data is reorganised for further processing. The next steps are performed on a per ship basis, and therefore the two files with the static and dynamic are combined and split per ship. This is supported by the application service "Split and combine", which is also provided by the application component "AIS data preparation".

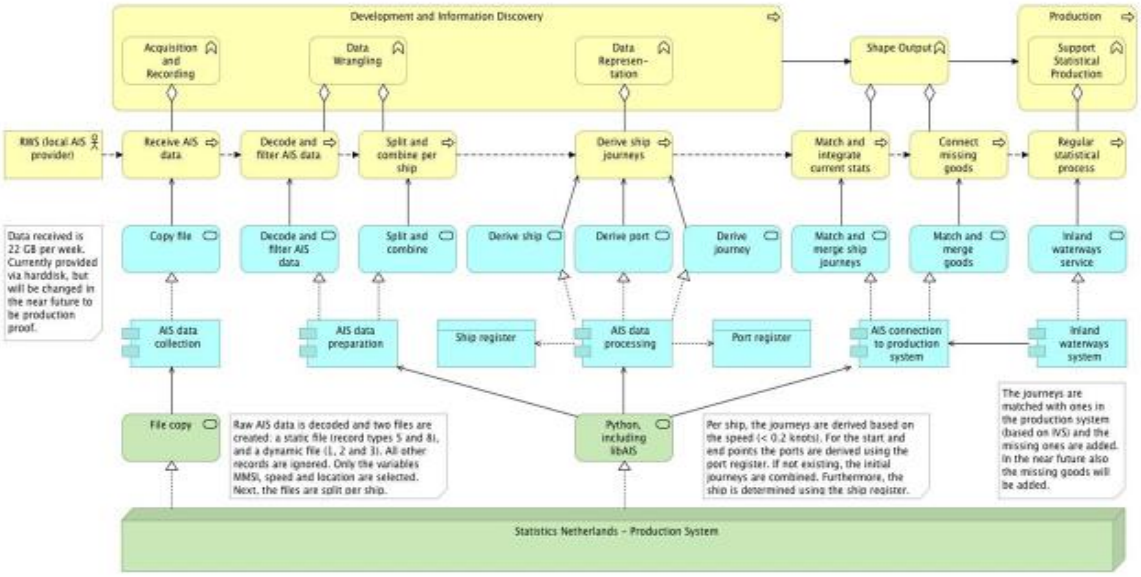


Figure 25 WPE Application Architecture

- Derive ship journeys

  This is the main part of the process. Here, each ship file is processed (can be performed in parallel) and the ship, the journeys, and the ports are derived. First, the ship is identified based on the MMSI, and all kind of characteristics are retrieved from the ship register. Then, the journeys are derived based on the speed. When the speed is below 0,2 knots, it is assumed that the ship has stopped at a port. These stops are checked using the port register. If it is not a port, because for example the ship is lying still for an overnight stay, the journeys are combined. At the end of the process, we know the ship, and all the journeys it made from which port to which port.

  This process is supported by the application services "Derive ship", "Derive journey" (including "Combine journey"), and "Derive port", all provided by the application component "AIS data processing". And the application component needs access to both the ship register and the port register.

- Match and integrate current statistics

  As mentioned, AIS is used to resolve the under-coverage in the current Dutch statistics. The inland waterway (IVS) production system already derives all ships and their journeys using other data sources. All journeys from AIS are compared with the ones in IVS and the missing ones are added. Match and merge seems a simple process, but linking is methodological challenging.

  This process is supported by the application service "Match and merge ship journeys", provided by the application component "AIS connection to production system", which has access to the IVS production system.

- Connect missing goods

  For the inland waterway statistics, we also need the goods that were transported. This information is not derived yet and therefore, this step is not yet implemented. Based on the depth of a ship, you might be able to derive the tonnage transported, and based on the type of ship and specifics of the port, you might be able to derived the type of goods transported. Whether this is possible, is still under investigation. If possible, an application service is needed to match and merge the goods in the IVS production system.

- Regular statistical process

  Now all data from AIS has been processes and merged into the IVS production system, the regular process can continue. This is business as usual.

Other AIS products

As mentioned, work package WPE actually consists of four subprojects and various products. Luckily, all these products can be created using a similar architecture. Of course, each product has its specifics, which need to be supported by the application and technical services.

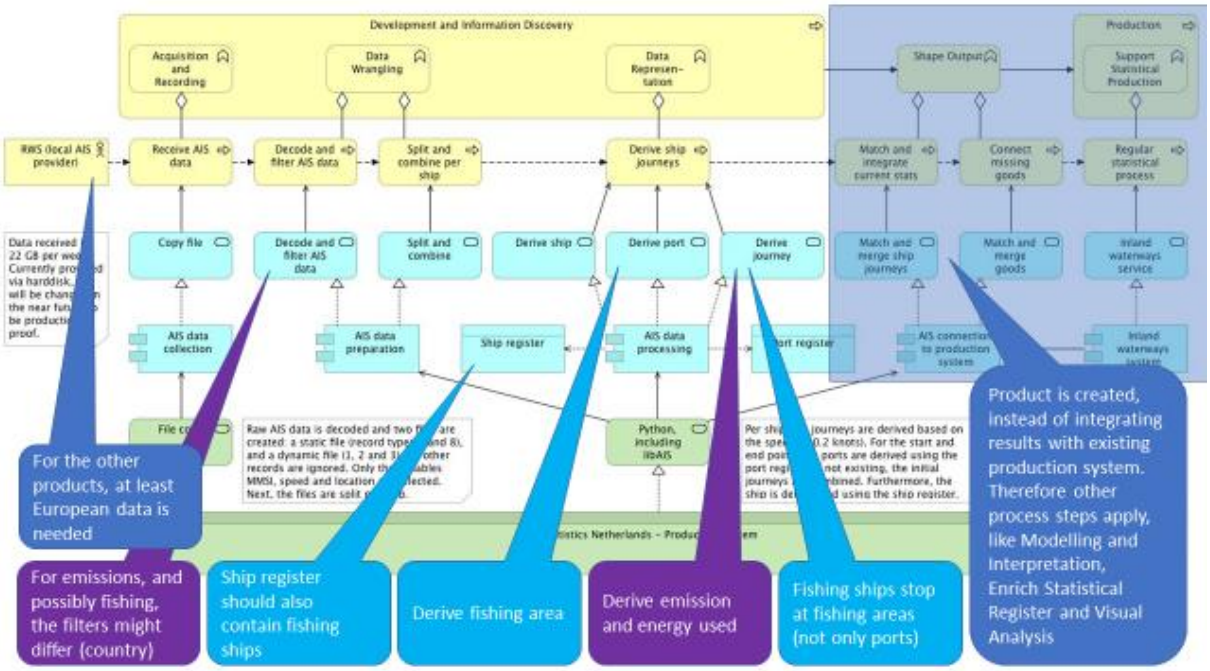


Figure 26 Other AIS products have similar architecture

In the picture above, the original architecture is shown with all changes necessary. Let's explain these per subproject:

- Port visits statistics

  For the port visits statistics, all port visits are already derived in the original architecture. Only change needed, is to actually create the mandatory F2 dataset using the information available.

  A different source might also be needed: national data could be used for a specific country, but European data or even worldwide data has a better coverage.

  If the port register is not available, it needs to be created, which is the business function "Enrich Statistical Register". And to analyse the data, already some beautiful animations are created, which are examples of the business function "Visual Analysis".

  The changes needed are shown in the picture in dark blue.

- Emissions and energy used

  In order to derive the emissions and energy used, some more changes are needed. Besides the source (worldwide coverage is needed) and the output (specific tables), we need two more changes. The changes needed are shown in the picture in purple.

  The first change, is that the filtering might differ. Emissions are calculated based on which country produced the emission. For ships, this is derived by the nationality of the ship (based on the flag or operating company). So, regardless where in the world the ship is sailing, the emissions belong to the country of ship's operating company nationality. So, in order to enhance the filtering, we might need to know the ship characteristics (see the application service "Derive ship") sooner.

  The second change, is the calculation of the emission and energy used. These should be calculated based on the mileage, speed, depth, type of engine, and many more variables. Possibly, even weather information is needed. Depending on the complexity, this could be part of the business function "Modelling and Interpretation"

- Fishing fleet statistics

  For the fishing fleet statistics, also some changes are needed, besides the source (at least European coverage is needed) and the output (specific tables). Of course, the ship register should also contain the fishing ships.

  The main difference, is that we need to know where the fishing areas are. Fishing ships catch their "cargo" at open sea. In the architecture shown, the ports were derived by the speed of ships (below 0,2 knots). Can fishing areas be derived likewise? So, the application services "Derive journey" and "Derive port" require some changes to also be able to identify fishing areas.

  Also, for the fishing fleet statistics, some interesting visuals are created to analyse the data, which are examples of the business function "Visual Analysis".

  The changes needed are shown in the picture in light blue.

Next steps are to detail the WPE application architectures for all four products, so that we can better compare these and determine the differences in more detail. It would also be interesting to compare these with the WPD application architecture, which is also on sensor data (smart meters), and for which we assume that there will be some similarities.

7. Overall conclusions

This deliverable describes the current status on the implementation and development of (experimental) statistical products to be delivered in this WPE, a quality review of AIS data itself from EMSA, the WPE architecture in BREAL and an overview of the data infrastructures used.

The implementation and development of all (experimental) statistical products are making progress:

(1) The implementation of AIS in the process of generating statistics on IWW is progressing. The next steps are to optimize processing-performance and to improve the algorithm matching journeys from AIS to journeys from the currently used (lock) data. Then, the next step is to estimate the type of goods ships might have carried, based on the type of terminal. Finally, consequences of the implementation on time-series will be investigated to determine if measures should be taken to obtain a consistent time-series.

(2) Develop statistics on the behaviour of fishing vessels: a ship register for the Polish fishing fleet was derived using the EU fishing fleet register. Using AIS, the first results on the activity of fishing fleet (time when a fishing vessel is out of the port) and the traffic of fishing fleet (number of fishing vessels in fishing areas) in Poland have been derived. Visuals have been created: a map with the fishing ports and fishing areas, and a map with traffic intensity in fishing areas. Future steps include (further) developing code on activity and traffic of the fishing fleet and the detection of fishing behaviour. Then, information about the weather conditions has to be added and visualization methods will be improved. Experimental statistics will be developed and the method tested for selected fishing ports in Greece and The Netherlands. Finally, a Starter Kit[4] will be developed.

(3) Improve quality of statistics on port visits: a process workflow has been developed using AIS data to derive the port visits and port traffic of the Piraeus port in Greece. So far, we focussed on the Piraeus Central Port where passenger, Ro/Ro passenger and cruise vessels arrive. The results are very encouraging. There are some difference we are going to investigate. Next, we will focus on Piraeus Cargo terminals for Cargo vessels and we will proceed with the compilation of F2 table. We are also going to put the code used on GitHub.

(4) Improve statistics on air emissions and energy used: Using text mining techniques, the ships' operators in the shipping list were matched with the shipping companies in the Dutch business register. More than 60% of the ships are not matched, so more research is needed. Next steps also include developing algorithms to calculate distances travelled on a global scale. Scenarios, taking into account more factors, like type of ship, will also be investigated.

Of course, all code and documentation will be made available.

Finally, the next steps described should all be taken in the next months and the results will be described in the final deliverable E4.

[4] A starter kit is a set of instructions and codes that will allow the user to prepare step-by-step, in this particular case, the experimental statistics on the fishing fleet.

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Annex 1: Scope of WPE in more detail

Work package number	**WPE**	Start Month: End Month:		**M1** **M24**
Title	**Tracking Ships**			
Partner (leader in bold) **Person-days**	**NL** **298**	EL 225	PL 170	

Objectives

Taking into account the developments of the successfully concluded work in the ESSnet big data programme during 2016-2018, the aim of this WP is to develop functional production prototypes including setting up procedures and developing technical solutions, to promote and support the collection, processing and analysis of (big) data from AIS (Automatic Identification System) for statistical production in the participating NSIs related to statistics on maritime, inland waterways and environmental statistics.

The output of the work should become available to the ESS and adequately enable other NSIs, in the next implementation phase, to put in place the necessary system(s) and deploy or adapt the proposed solution(s) in order to produce statistical output. From the practical point of view, the output should in addition develop key artefact like target architecture definition and requirements and guidance for the next implementation and deployment phase, hence producing specifications for full-fledged implementation at the ESS level. Therefore, the active involvement of subject matter experts, statisticians, enterprise architects and other stakeholders (for example Working Groups of Inland waterways and maritime statistics, Transport and Environment Reporting Mechanism (TERM) and Task Forces) is important. Subject matter experts for the NSIs working in the relevant national statistical production units will be involved in all phases of work. Participation to relevant Eurostat Working groups and Task Forces will be part of the work of this WP. This includes presentation of findings and work to the relevant working groups at an early stage.

Along the production of the relevant methodologies, recommendations, specifications and statistical software, the production of experimental statistics demonstrating the capabilities to produce statistics is a prominent objective.

Methodological, quality and technical results of this WP, including intermediate findings, will be used as inputs for WPF and WPK. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WPA.

Description of work

With respect to the development and implementation of functional production prototypes according to the work carried out during 2016-2018 by the preceding ESSnet, this WP aims at completing the following activities that would lead to the implementation and integration of AIS data within the national statistical production process(es) in the NSIs:

Task 1 - Interaction with WPF (NL)

Aim of this task is coordinating the interaction with WPF. The aim of WPF is the definition of reference architectures necessary to carry out big data production both at national and European levels. This task

includes (i) designing and adopting application and information architectures (WPF Task 1) and (ii) providing input to WPF Task 2.1 on solution architectures.

Task 2 - Data access (NL)

AIS are internationally standardized encoded data available for national territories and the entire European territory, as well as at global level. The used European data source in ESSnet big data programme 2016 -2018 isn't recent enough to use for implementation.

This WP aims to implement AIS in maritime statistics as well as in inland waterway s statistics. That is why it is important to have European AIS data on maritime ships available as well as national data on Inland waterway ships. The European Maritime Safety Agency (EMSA) was requested for European AIS data from 2017 and 2018 on Maritime ships. Dutch AIS data on inland ships will be provided by Rijkswaterstaat for implementing AIS in the statistical process of inland waterways.

This task aims to:

- Collect the data on maritime ships at a European level and collect the national data on Inland waterway ships for the Netherlands
- Set a sustainable long term collaboration frame with EMSA for future use at European level. This will require clear roles related to data, pre-processing, processing and production of statistics. Preparation of this future use of European AIS data from EMSA is also part of this task.

For this project it is essential to get at least access to European AIS data (from EMSA). At this point it seems that EMSA could provide the data needed for this project, but there is no fina l agreement yet. Costs for buying European AIS data from a commercial organisation, in the case EMSA cannot provide the data, are not budgeted yet.

Task 3 - Definition (EL, NL, PL)

Aim of this task is to define the specific outcome of this WP. What would be the delivered statistical product (both new statistics and enhanced existing statistics) at the end of this ESSnet? In what statistical processes AIS would be implemented? How would subject matter experts, statisticians, enterprise architects and other stakeholders (Working Groups of inland waterways and maritime statistics and Task Forces) be involved?

Key elements of this task are:

- Definition of statistical products/prototypes. Including a plan for possibilities for testing these prototypes in other countries.
- Identification of statistical production processes and capabilities that may be affected at national level
- Guideline for involving subject matter experts, statisticians, enterprise architects and other stakeholders. This includes involving subject matter experts for the NSIs working in the relevant national statistical production units in all phases of work as well as participation to relevant Eurostat Working groups and Task Forces. EMSA will follow the progress during this project. EMSA is also available for (methodological) advice.
- Description of an adequate Reference Methodological Framework for processing AIS data in statistical processes. Setting a sustainable long term collaboration frame with EMSA for future use

of AIS data at European level will be included in building the necessary reference architecture using European data.

- Development of data models (output in XML or other standard), data verification and data exchange protocols for all the selected statistical processes for implementation

Task 4 - Prototypes (EL, NL, PL)

The aim of this task is the development of fully functional prototypes of the identified statistical production processes in task 3. Including a description on the adaptation and consolidation of the methodology, the procedures and the tools for collecting. Output would include software as well. This task also includes a definition of:

- the implementation requirements (report) of prototypes in the relevant statistical production processes at European and national level (all)
- a quality management framework (report) to ensure quality of outputs and process, leading to complementary statistics (indicators) (EL, NL)
- required metadata throughout the process (report, metadata schema) (all)
- a risk plan with mitigation scenarios (all)

Testing of the prototypes (software, architecture, technical requirements, etc.) will take place (if possible) in (at least) the participating countries of this WP. This testing phase would then allow corrections or development of alternatives to be adapted to national specificities of (at least) the partners in this WP.

Besides this, there will be taken into account a couple of recommendations resulting from the previous ESSnet on big data. These two topics will be further investigated in this ESSnet:

- Combining AIS with online information and/or information on IHS Maritime to provide more reliable information on the type and capacity of ships (NL, EL).
- Developing an algorithm for signalling anomalies in the movements of ships (e.g. different traffic intensities or travelling speeds) to get information on disruptions or (changed) regulations in ports and fairways (NL, PL).

The outcomes of the two above mentioned investigations will be really helpful for further implementation of AIS in official statistics in this ESSnet.

The three topics beneath will be further investigated in this WP as well:

- Developing an algorithm that can construct a reference frame of ports. The next step would be to construct a reference frame of terminals. Combining this with type of visiting ship could assist with ship visits where type of ships or goods is missing. Now ports and terminals have to be defined manually, this is not feasible for countries with many ports (NL).
- Improving the port to port distance matrix by Investigating more journeys and factors. Optimizing algorithms could also result in using actual AIS journey data instead of the average distance matrix. Optimizing algorithms could also result in using actual AIS journey data instead of the average distance matrix (NL).

- Using AIS as a fast indicator of international trade, where international comparisons can be made quickly. National offices are really interested in fast indicators of economic development (NL)

On some of these topics there are already some first findings available, resulting from the ESSnet big data programme 2016-2018. It is not foreseen that the outcomes of these investigations will directly be implemented in one of the statistical offices during this ESSnet, but they will provide very valuable insights for the ESSnet as a whole.

The results of all investigations mentioned will be produced as experimental statistics on Eurostat's dedicated section (task 4).

Task 5 - Experimental statistics (EL, NL, PL)

The aim of this task is to produce experimental statistics by visualising the first results of the outcomes on the prototypes in task 4. The experimental statistics will be published most likely in the form of visualisations and they will be hosted by Eurostat's dedicated section and on a national level.

Task 6 - Implementation (EL, NL, PL)

This task involves (the preparation of) implementing AIS in statistical processes. After finishing this task there will be at least:

- A description (process model) on the future process of all selected processes in task 2, describing the processes including tools, infrastructure and conditions (for example skills needed, hardware requirements).
- A report describing the activities needed for coming from the experimental statistic to the future process, for all selected processes in task 2. What needs to be done before the future process is ready for use and regular output can be published?
- One statistical process where AIS is fully implemented, including:

1. Production of tools and software (source code) that can be applied by other NSI's
2. Description of used methodologies
3. A developer's and user's handbook
4. A procedure for testing and maintenance.

Besides the above mentioned deliverables, the following more general actions will be performed in this task:

- Address issues related to sustainability of data sources, data use by NSIs, as well as data sharing between NSIs (apply risk management principles).
- Develop and evaluate scenarios for data governance and data management in the wider aspect of data sharing and collaboration processes management.

Task 7 - Future perspectives (EL, NL, PL)

Aim of this task is to summarise the project results in a final detailed report with the outcomes of this WP. Guidelines and documented material for full-fledged implementation in the ESS including relevant software. Setting a sustainable long term collaboration frame with EMSA for future use at European level will require

clear roles related to data, pre-processing, processing and production of statistics. Preparation of this future use of European AIS data from EMSA will be will be part of this final report. The results will also include an assessment of sustainability over time.

Deliverables

E1 (M7)

- Report on the results of the definition process, concerning the Reference Methodological Framework and data models (results of task 3).

E2 (M12)

- Report on implementation requirements of relevant statistical production processes at European and national level, quality framework, required metadata and a risk plan (results of task 4).

E3 (M16)

- Interim technical report concerning the conditions for using the data, the methodology and the procedures to be used for producing statistics (development of functional prototypes, development toolkit and software), recommendations and guidelines for full-fledged implementation in the ESS

E4 (M24)

- Final detailed report with the outcomes of this WP. Guidelines and documented material for full-fledged implementation in the ESS including relevant software, description of used methodologies, developer's and user's handbook, procedure for testing and maintenance and description of skills needed. The results should also include an assessment of sustainability over time.

Milestones

EM1 (M9)

- Report on the WP meeting mid 2019

EM2 (M20)

- Report on the WP meeting mid 2020

Annex 2: Variables Shipping List

LRNO
DOC Company
DOC Company Code
Ship Trading Status
Ship Type
Vessel Sub Type
Sector NL
Year of Build
Current Name
Current Name Effective Date
Last Former Name
Last Former Name Effective Date
Registered Owner
Registered Owner Code
Registered Owner Country of Residence
Registered Owner True Nationality
Dutch ownership
Ship Manager
Ship Manager Code
Ship Manager Country of Residence
Ship Manager True Nationality
Operator From Current File
Operator Code From Current File
Operator Country of Residence from Current File
Operator True Nationality from Current File
Group Beneficial Owner
Group Beneficial Owner Country of Residence
Group Beneficial Owner True Nationality
Group Beneficial Owner Country of Domicile
Flag Decode
Flag Effective Date from Current File
Previous Flag From Current File
Previous Flag Effective Date From Current File
GT
Dwt
NT
PS_BEID

Annex 3: Quality framework in more detail

Hyper dimension	Quality Dimension	Possible indicators	Results	Conclusion
Source	*Institutional/ Business Environment*	1. What is your estimate of the overall risk that the data provider will not meet the quality requirements of the NSO?	Probability: Medium, Impact: Medium	Under investigation OK
		2. What is the risk that the BDS will not be available from the data provider in the future? If it will not, will there be comparable data sources in the future?	Probability: medium, Impact: high (no other provider)	NOT OK
		3. How relevant are the data, if they would be available for only a short period of time?	Not very relevant	OK
		4. How long do these data need to be available to be relevant?	Minimally two years, to be able to filter out seasonal effects and see developments	OK
		5. Is it likely that we can replace these data with similar (or next generation) data, once the data source or technology becomes obsolete?	Yes, new technology will still contain information needed.	OK
	Privacy and Security	1. Does the NSO have clear legal authority to obtain the data?	yes	OK
		2. Are there legal limitations or restrictions on the use to which the data can be put?	No	Action needed: solve privacy issues
		3. Are the data provider and the NSO willing to enter negotiations to solve any legal issues, if necessary?	yes	OK
		4. Was the data collected in accordance with relevant privacy laws?	yes	OK
		5. Do the NSO's own confidentiality policies limit the utility of data?	no	OK
		6. Are stakeholders (private sector, public, others) likely to react negatively given the intended use of the data by the NSO?	no	OK
		7. Will there be a need to carry out privacy assessment exercises and public consultations in relation to using this data and its potential impact on the NSO reputation and credibility?	Only if it would be used for IWW, but it will not be used for that.	OK
Metadata	*Complexity*	1. Structure: how easy would it be to render the data source into a useable structure (i.e… one record per unit of observation,)?	easy	OK

		2. Format: Is the data source in standard format (e.g., XLS, XML)? How many different formats were used in the data source? How easy would it be to render the data source variables into a useable format (i.e... parsing, grooming, coding, treatment of outliers or missing values)?	no, only in NMEA format, easy	OK
		3. Data: how many different standards were used in the data source (e.g., ISO-3166 to describe countries)? Is there any non-standard code lists used in the data source that are not unified? How many different code lists were used in the data source?	NMEA itself is an ISO standard	OK
		4. Hierarchies: is there a hierarchical relationship between records or variables?	no	OK
		5. Structure: How many different files or tables are in the data source?	one	OK
	Completeness	1. Qualitative assessment (e.g. score for completeness of metadata for input phase: 0 description missing, 1 description insufficient, 2 description complete)	2	OK
		2. In case of missing/incomplete descriptions what are the consequences/drawbacks for data usability?	Not applicable	OK
		3. Are the population units defined clearly?	yes	OK
		4. Are the variables defined clearly?	Yes	OK
		5. Qualitative assessment of completeness and clarity of metadata	Metadata is not really clear on coverage and filtering	t OK, because we know from EMSA how the filtering works
		6. In case of unclear/ambiguous descriptions what are the consequences/drawbacks for data usability?	Can be derived from the data	OK
	Usability	1. Will the NSO need to acquire new skills to use and analyse the data?	Yes, big data skills	OK
		2. How much resourcing will cleaning and processing the dataset require?	Processing (splitting in static and dynamic messages) takes about 20 minutes for 6 months of data	OK
		3. How big is the data set?	150 GB (encoded)	OK
		4. Data transmission: are special arrangements for data transmission required, and if so, can the NSO meet those requirements?	Yes, for getting the data at once for 2 years? No, when receiving the data in small, daily portions, the	OK

			transmission should not be an issue	
		5. IT requirements: what would be the hardware and software requirements to process and store this data? Will there be a need to develop a specific IT infrastructure?	Yes, Big Data infrastructure (i.e., Hadoop, Spark)	OK
	Time-related factors	1. Time between receipt of data and when the data was collected;	Not applicable	OK
		2. When was the data collected? What is the reference period of the data?	January 2016 – December 2017	OK
		3. Whether data is collected and available periodically. Recurring data provides the opportunity for time series.	Data is available continuously, but we only obtained two years of data	OK
		4. Could changes in concepts or methods limit the potential use of historical data?	No, methods are developed to process long historical time series.	OK
	Linkability	1. Are potential linking variables present on the file that could be used for data integration with other data files?	Yes, the IMO number is available. This number is a unique number for sea vessels.	OK
		2. Calculate the percentage of units linked and not linked in both the Big Data (BD) and other data sources. The indicator is the percentage of units linked unambiguously (strong link) / percentage of units linked with a soft link (linking requirements were relaxed in order to link more units)	This is done by checking the AIS source with other sources and AIS sources with each other	OK
	Coherence - consistency	1. How do you rate the variables capturing the constructs that are of interest?	High	OK
		2. Are the definitions used aligned with NSO standards?	yes, data is at such a fine grain level that it is easy to create aggregates that are aligned	OK
		3. Do the anomalies in the data indicate important errors that would limit the potential use?	no	OK
	Validity	1. Is the metadata available sufficient to assess the soundness of the methods used?	It is insufficient. We do not know the filtering used. Filtering weten we nog niet	Not OK
		2. Are there critical flaws in the processes that would limit potential use of the data?	No, despite some variables are not filled out consistently	OK

			by the owner of the vessel	
Data	*Accuracy and selectivity*	1. If a reference data set is available, assess coverage error. For example, measures of distance between Big Data population and the target population (e.g. Kolmogorov-Smirnov Index, Index of dissimilarity)	see deliverable 4.3 [5]	OK
		2. Does the file contain duplicates?	No	OK
		3. Are the data values within the acceptable range?	Yes	OK
		4. Assessment (also qualitative) of sub-populations that are known to be under/over-represented or totally excluded by Big Data source.	Little vessels do not have AIS on board. Because we were only interested in maritime ships this was not a problem. For fishing ships, the data does not cover all ships.	OK
		5. Assessment of spatial distribution of measurement instrument and of periodicity of observations	Not applicable	OK
	Linkability	1. Are potential linking variables present on the file that could be used for data integration with other data files?	yes, IMO for linking vessels and Lat/Lon for linking positions to ports	OK
		2. Calculate the percentage of units linked and not linked in both the Big Data (BD) and other data sources. The indicator is the percentage of units linked unambiguously (strong link) / percentage of units linked with a soft link (linking requirements were relaxed in order to link more units)	see deliverable 4.2 [4] and 4.3 [5]	OK
	Coherence - consistency	1. How do you rate the variables capturing the constructs that are of interest?	high	OK
		2. Are the definitions used aligned with NSO standards?	yes	OK
		3. Do the anomalies in the data indicate important errors that would limit the potential use?	no	OK
	Validity	1. Is the metadata available sufficient to assess the soundness of the methods used?	It is insufficient. We do not know the filtering used.	Not OK
		2. Are there critical flaws in the processes that would limit potential use of the data?	No, despite some variables are not filled out consistently by the owner of the vessel	OK